

Benchmarking and Performance Analysis of LLM Inferencing Systems

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Systems for LLMs
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1 Radix Tree Workload Modes

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Full Sharing (All share 3L)

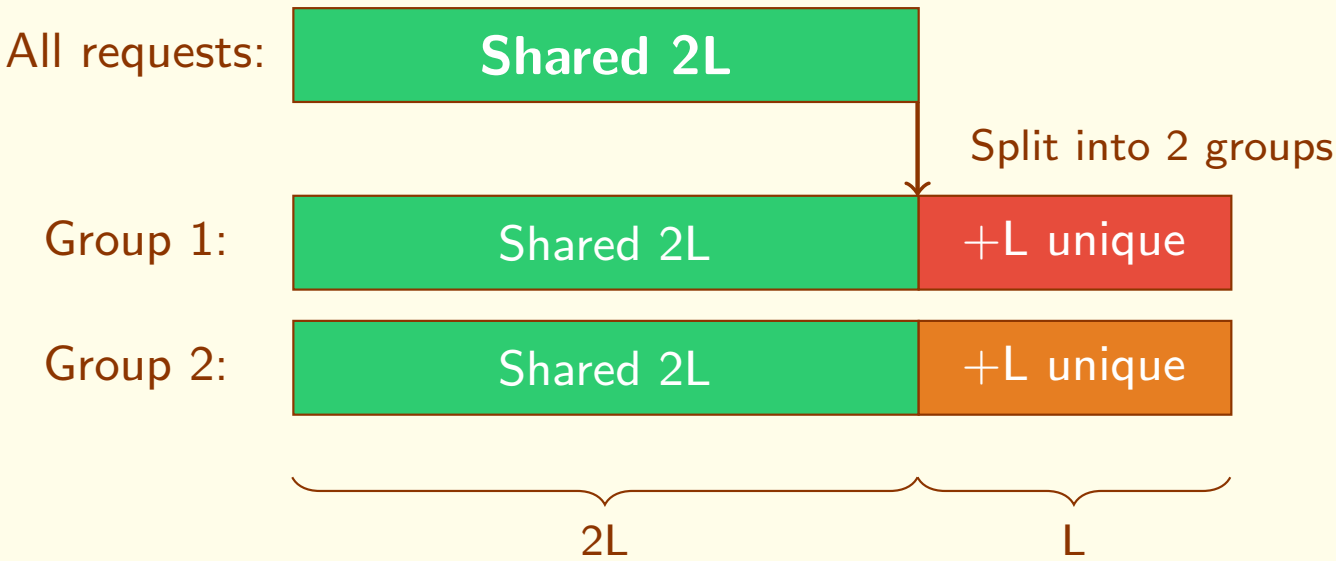
All requests: **Shared 3L**

3L tokens

Mode 2



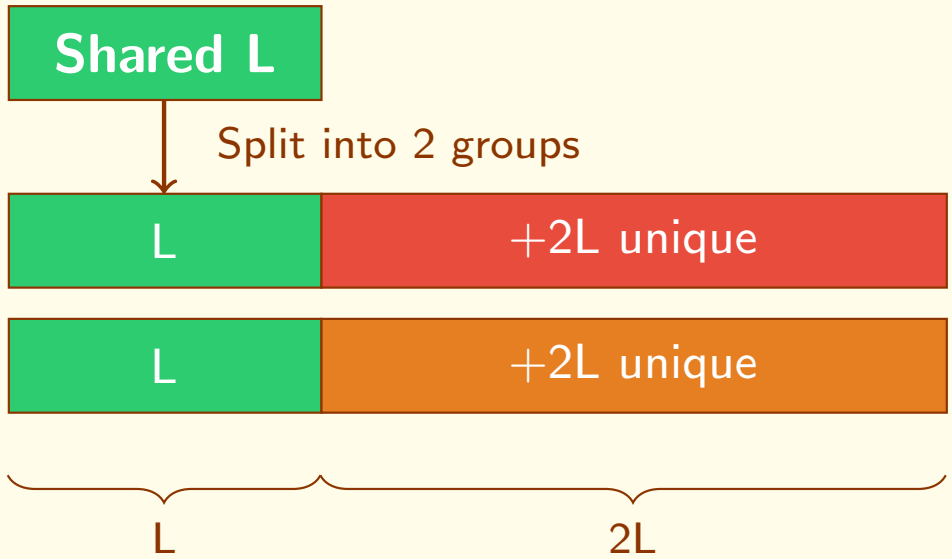
All share 2L, then 2xL





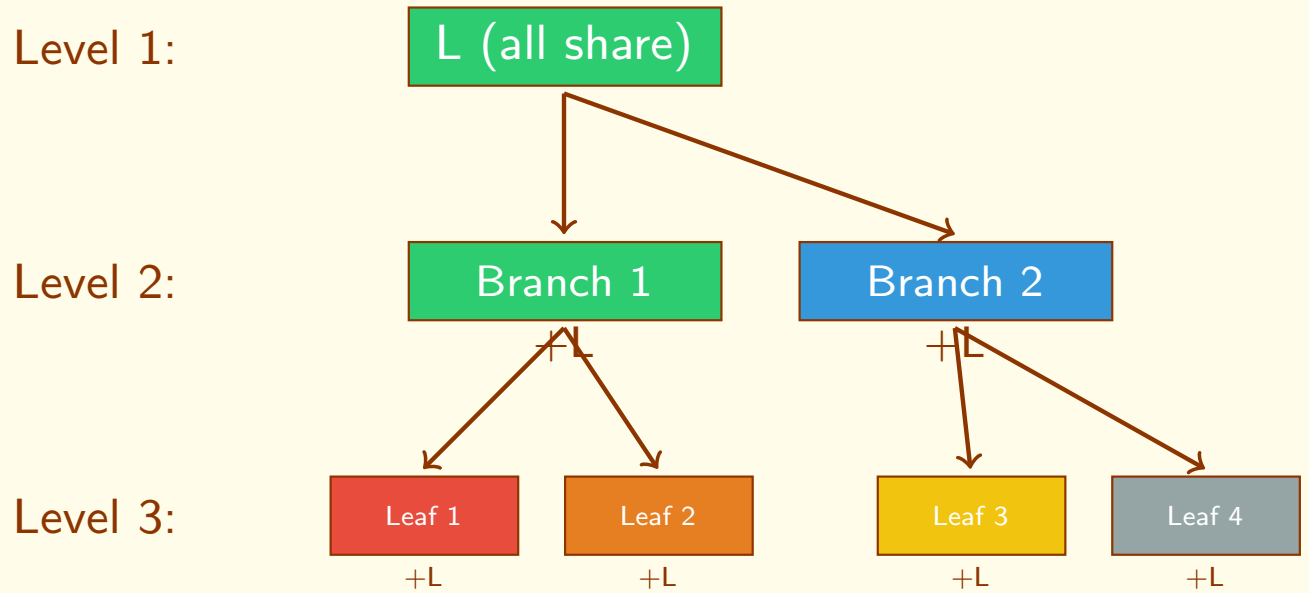
All share L, then 2x2L

All requests:





Three-Level Tree (L → 2L → 3L, 4 leaves)



Mode 5



2 separate groups with 3L each

Group 1: **Context A (3L tokens)**

Group 2: **Context B (3L tokens)**

3L tokens (completely different)

Results across varying L

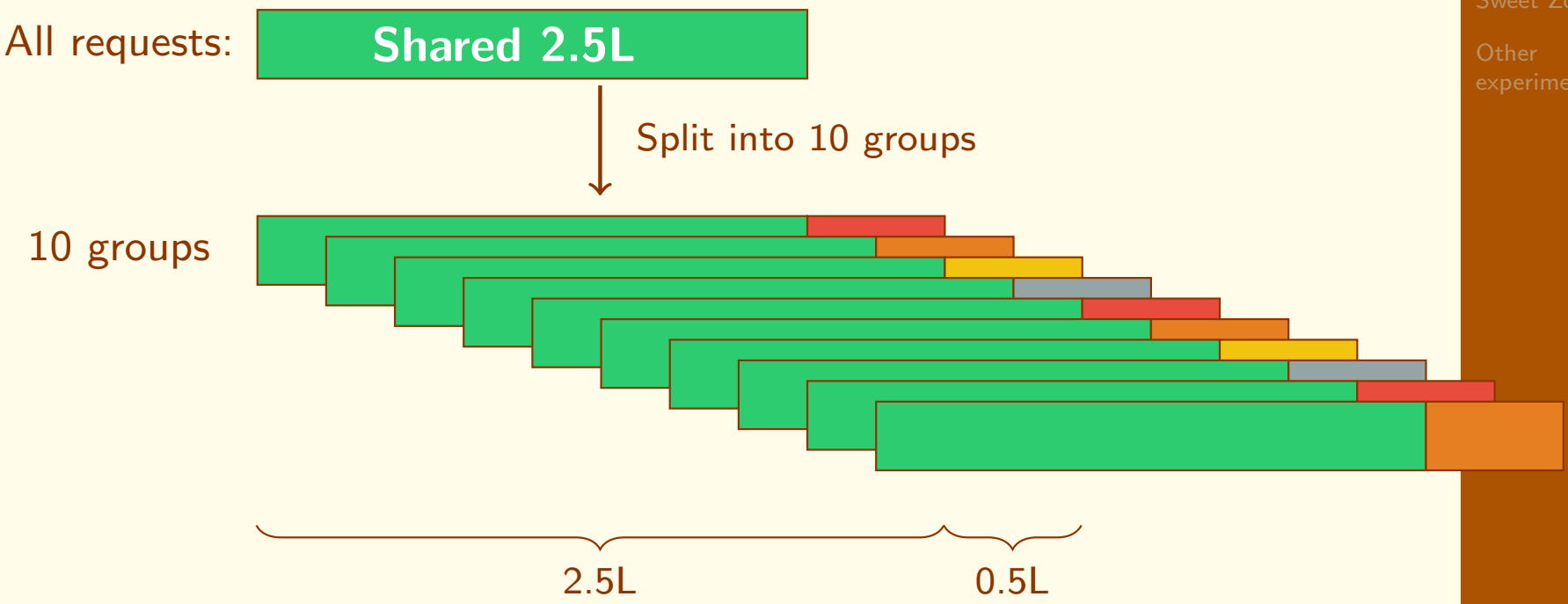


Figure: Results across varying L

Mode 6



2.5L shared + 10x0.5L



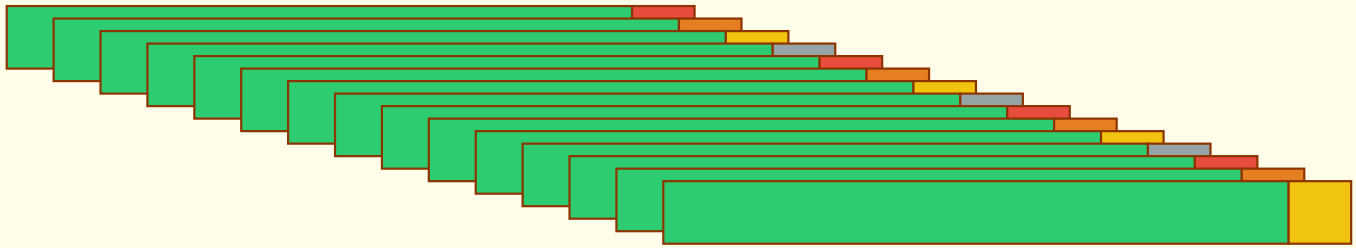
Mode 7



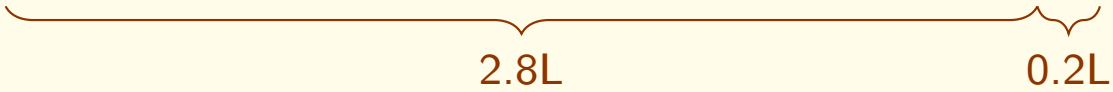
2.8L shared + 50x0.2L

All requests: **Shared 2.8L**

50 groups



...(More groups)



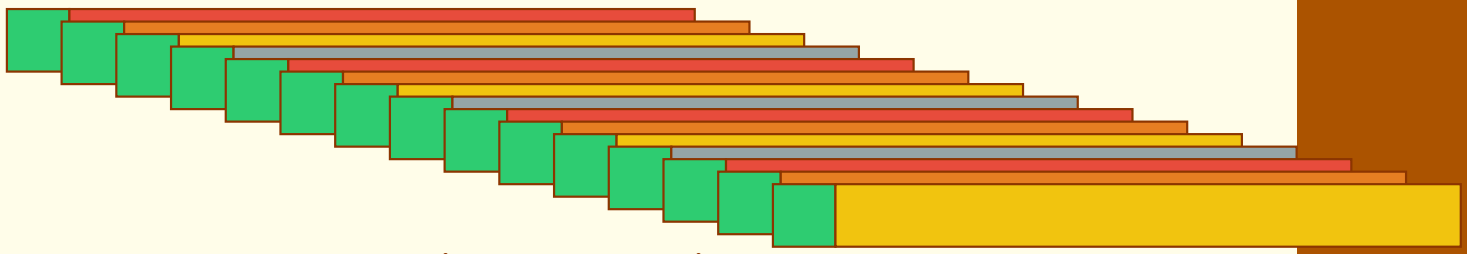
Mode 8



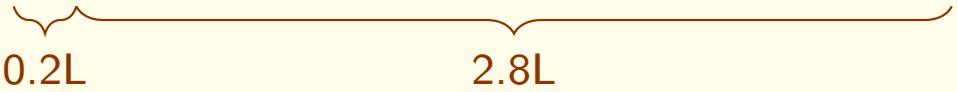
0.2L shared + 50x2.8L

All requests: 0.2L

50 groups



...(More groups)



Comparison of Throughput across different modes



Modes	**L = 1000**	**L = 2000**
Mode 1 - All share 3L	29.18	27.97
Mode 2 - All share 2L, then 2xL	19.94	14.06
Mode 3 - All share L, then 2x2L	15.40	8.89
Mode 4 - L $\rightarrow$ 2L $\rightarrow$ 3L, 4 leaves	15.41	9.88
Mode 5 - 2 separate groups with 3L each	10.18	6.62
Mode 6 - 2.5L shared + 10x0.5L	25.34	20.27
Mode 7 - 2.8L shared + 50x0.2L	28.05	23.17
Mode 8 - 0.2L shared + 50x2.8L	10.05	6.48

Table: Comparison of Throughput across different modes

How many families?



- A high request arrival rate (500 req/sec)
- Max batch size of 500
- A shared prefix of length 6500 tokens
- 100 new output tokens
- Model: Llama3.1-8B

Number of Families	vLLM Baseline (FCFS)		vLLM (User-provided Family IDs)	
	Average Batch Size	Throughput	Average Batch Size	Throughput
10	465.48	5.92	141.23	21.94
20	455.47	5.74	72.67	15.27
50	57.48	2.07	64.77	14.84
100	41.97	1.11	67.14	14.82

How many req/sec?



- 5 families
- Max batch size of 500
- A shared prefix of length 6500 tokens
- 100 new output tokens
- Model: Llama3.1-8B

req/sec	vLLM Baseline (FCFS)		vLLM (User-provided Family IDs)	
	Average Batch Size	Throughput	Average Batch Size	Throughput
1	2.6	1.04	2.02	0.95
2	5.15	2.00	4.48	1.99
5	29.60	5.09	11.11	4.40
10	395.80	4.42	23.96	8.16
20	438.85	5.25	52.79	13.32

How long should the shared prefix be?



- A high request arrival rate (500 req/sec)
- Max batch size of 500
- 5 families
- 100 new output tokens
- Model: Llama3.1-8B

Shared Prefix Length in Number of Tokens	vLLM Baseline (FCFS)		vLLM (User-provided Family IDs)	
	Average Batch Size	Throughput	Average Batch Size	Throughput
13	455.48	42.47	141.23	44.67
65	448.87	40.14	318.60	43.96
130	460.09	38.19	330.49	41.99
260	462.80	35.34	326.36	44.58
650	469.12	27.47	335.21	42.00

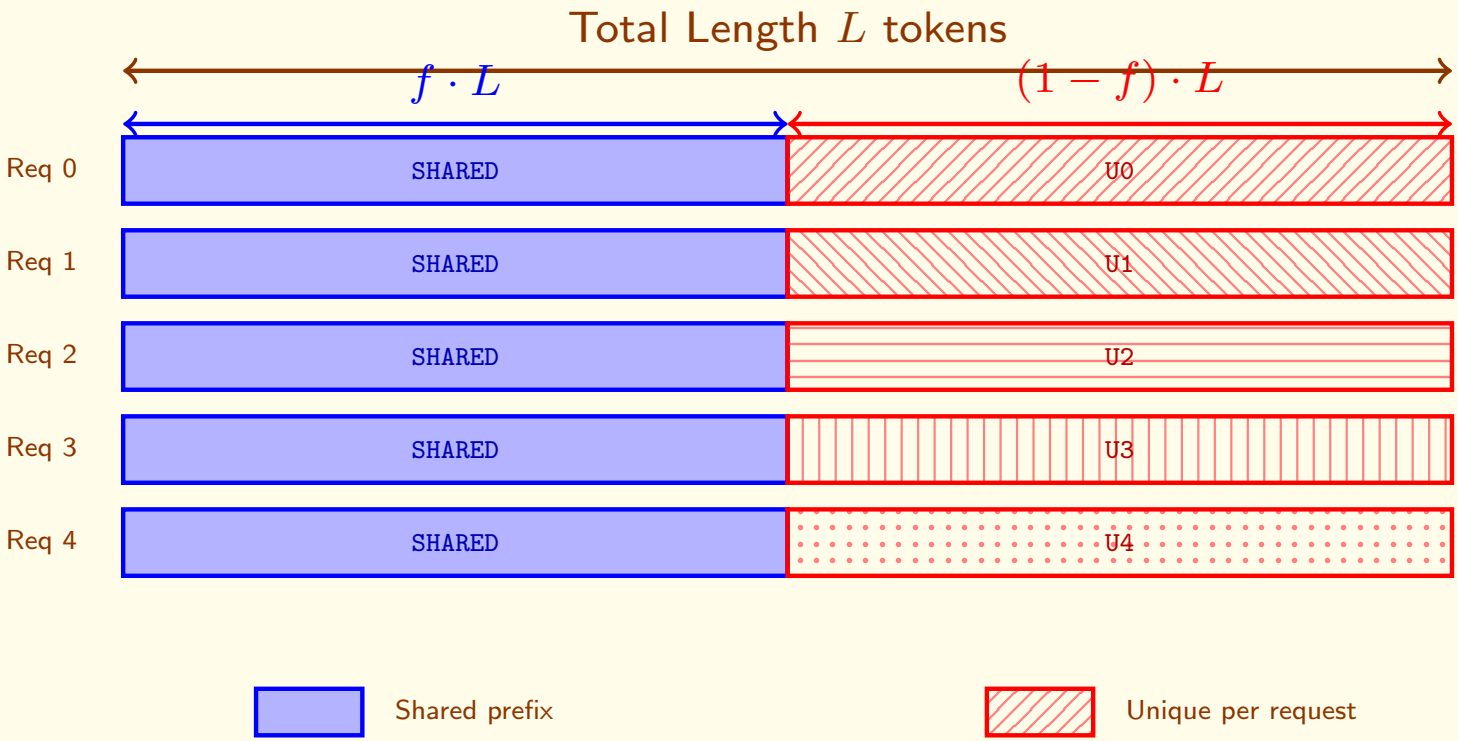
How large should the model be?



- A high request arrival rate (500 req/sec)
- Max batch size of 500
- 5 families
- 100 new output tokens
- Shared prefix length: 6500 tokens

Model Name	vLLM Baseline (FCFS)		vLLM (User-provided Family IDs)	
	Average Batch Size	Throughput	Average Batch Size	Throughput
Qwen2.5-0.5B	378.98	25.40	89.63	49.08
Llama3.1-8B	448.87	4.34	151.21	30.84
LongChat-13B	5.28	0.61	215.70	22.01

Varying the amount of sharing



Varying the amount of sharing (L = 100 tokens)

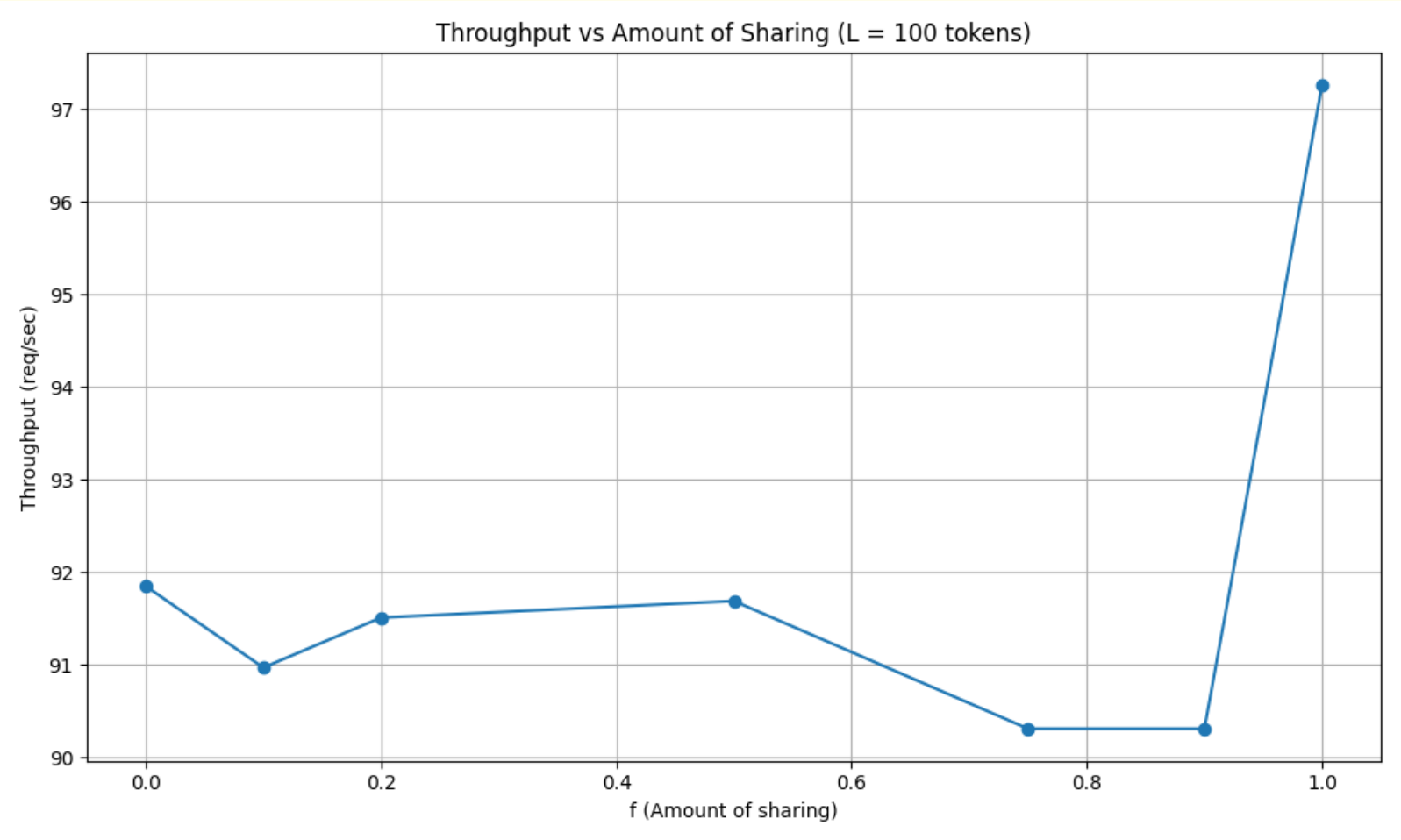


Figure: Results across varying L

Varying the amount of sharing (L = 500 tokens)

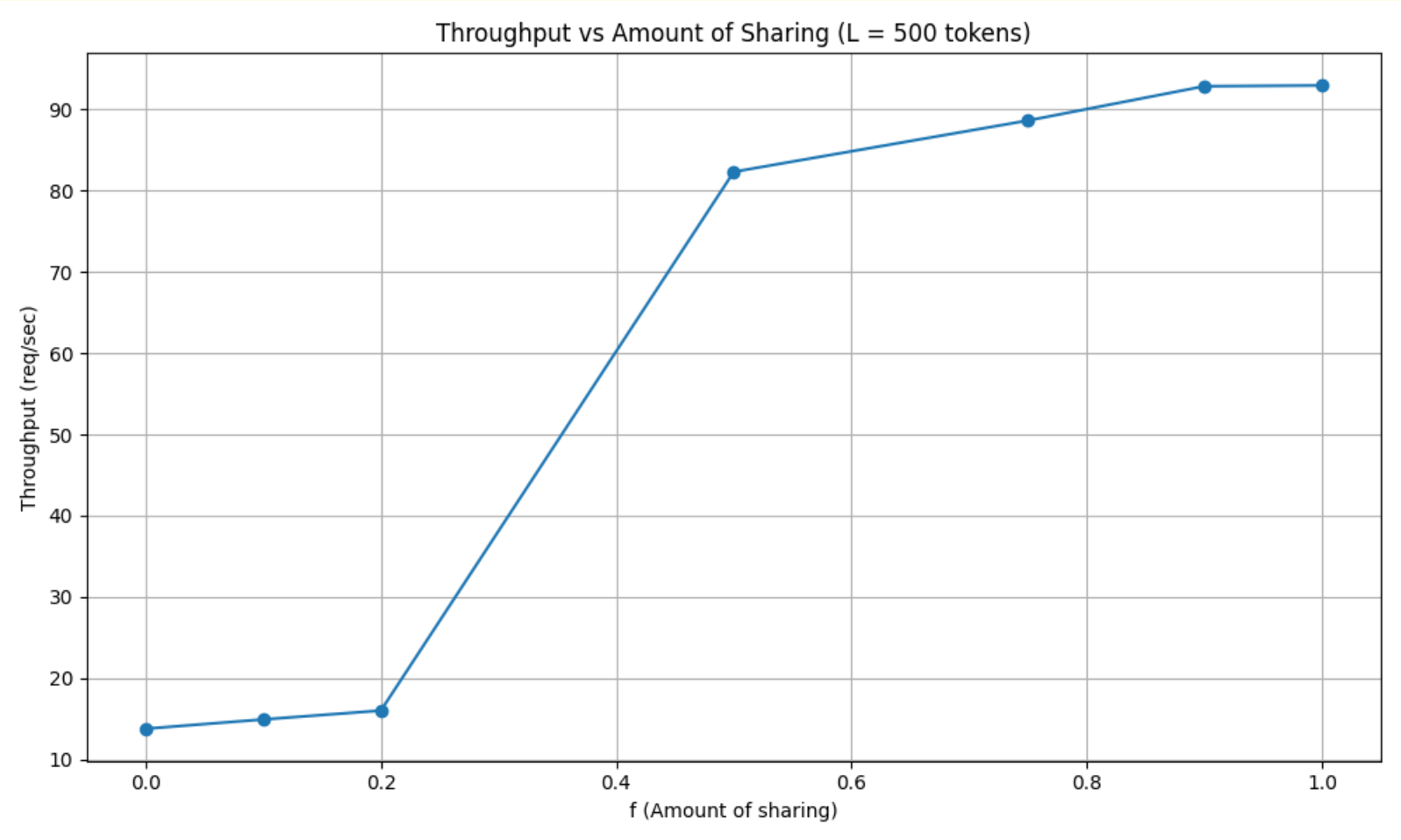


Figure: Results across varying L

Varying the amount of sharing (L = 1000 tokens)

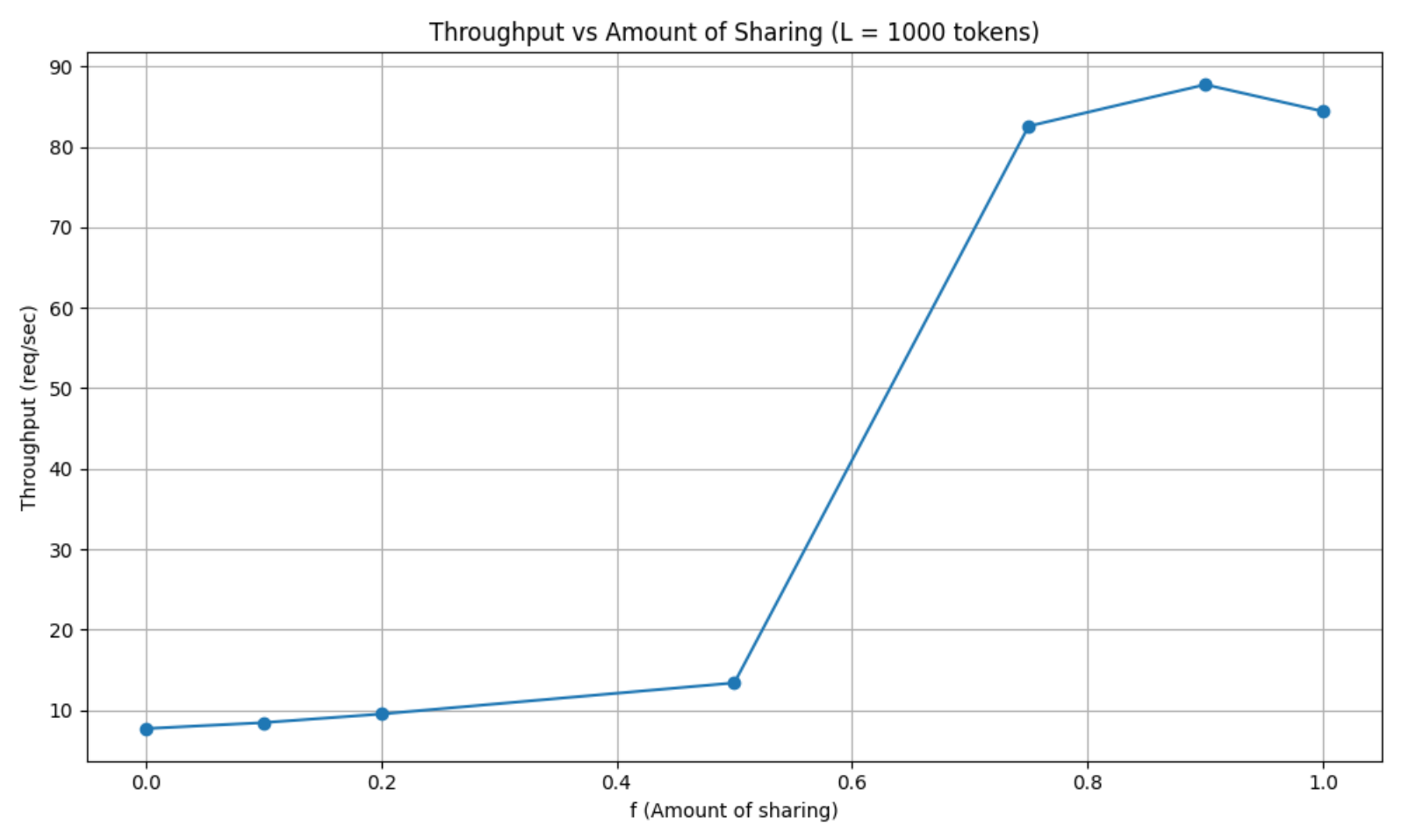


Figure: Results across varying L

Varying the amount of sharing (L = 2000 tokens)

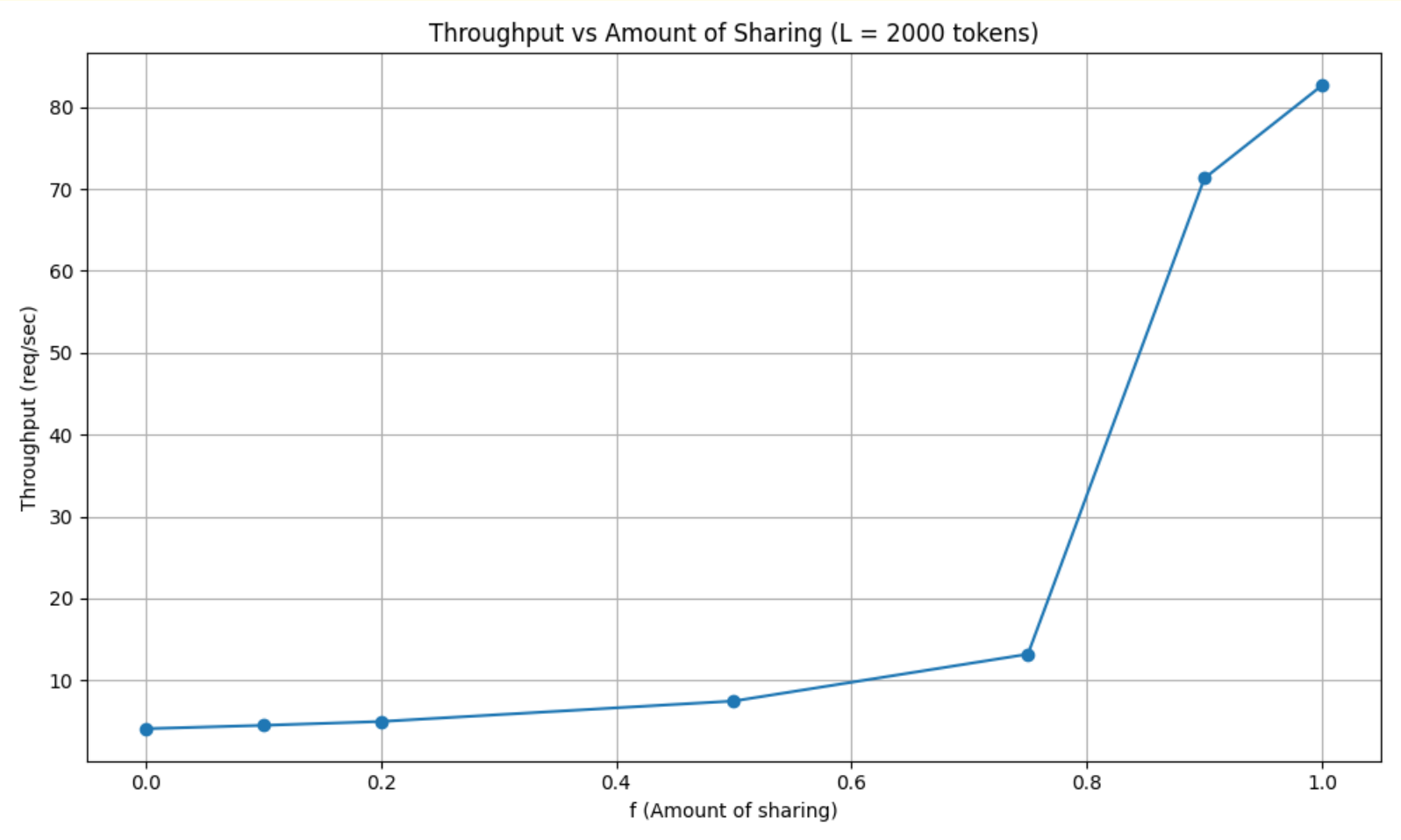


Figure: Results across varying L

Varying the amount of sharing (L = 500 tokens - More points)

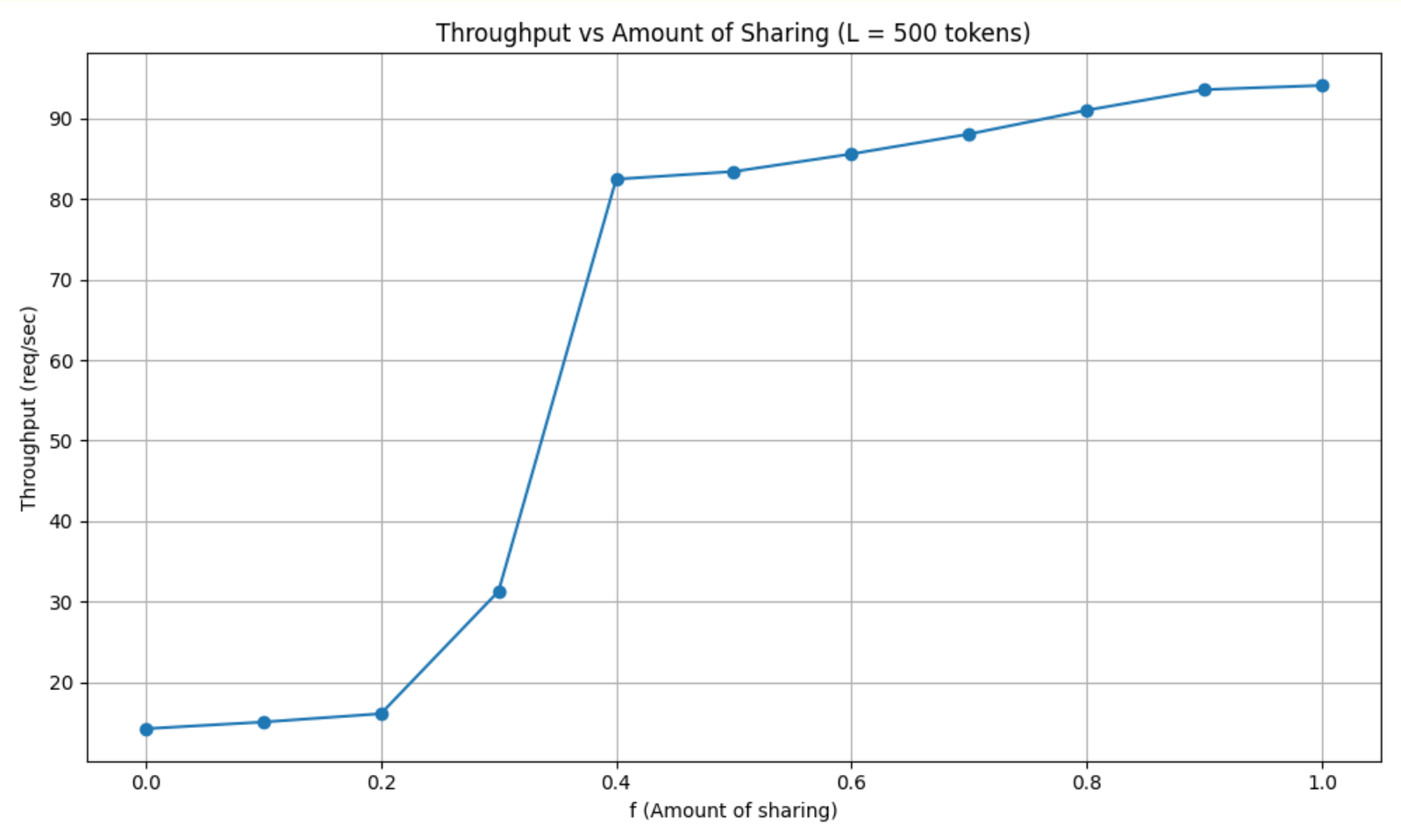
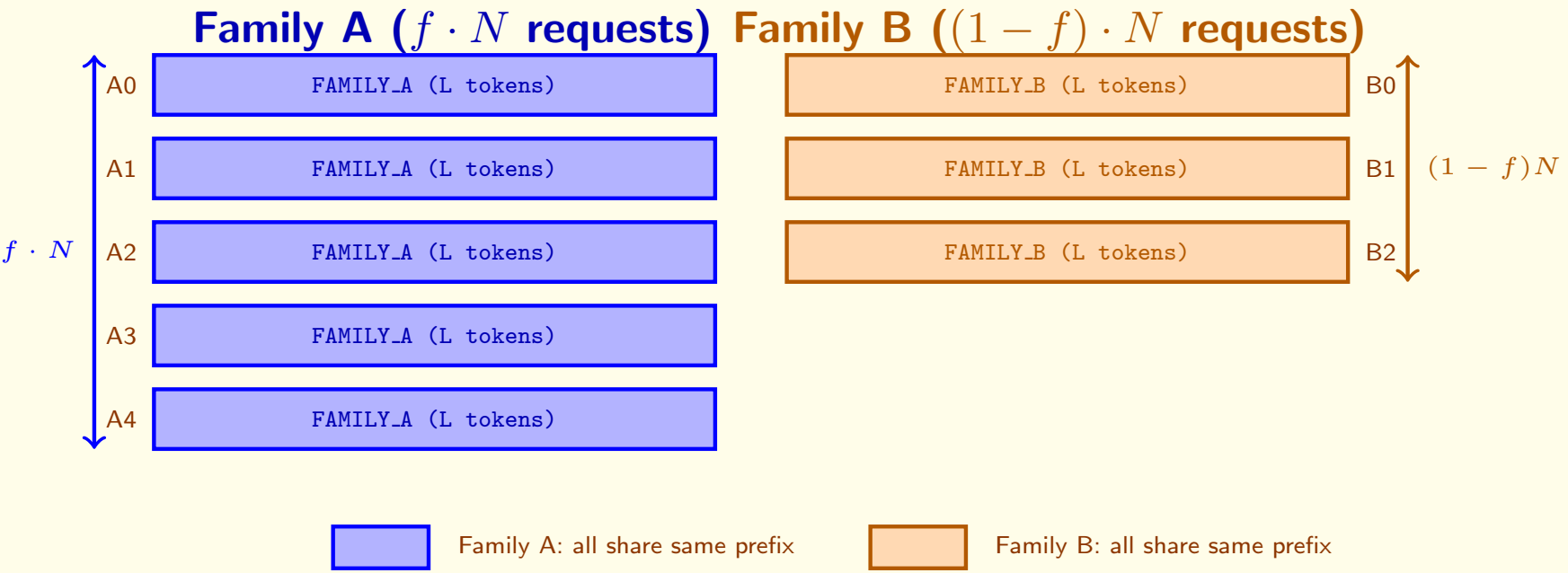


Figure: Results across varying L

Two-family batch mixture



Two Family Batch Mixture (L = 100 tokens)

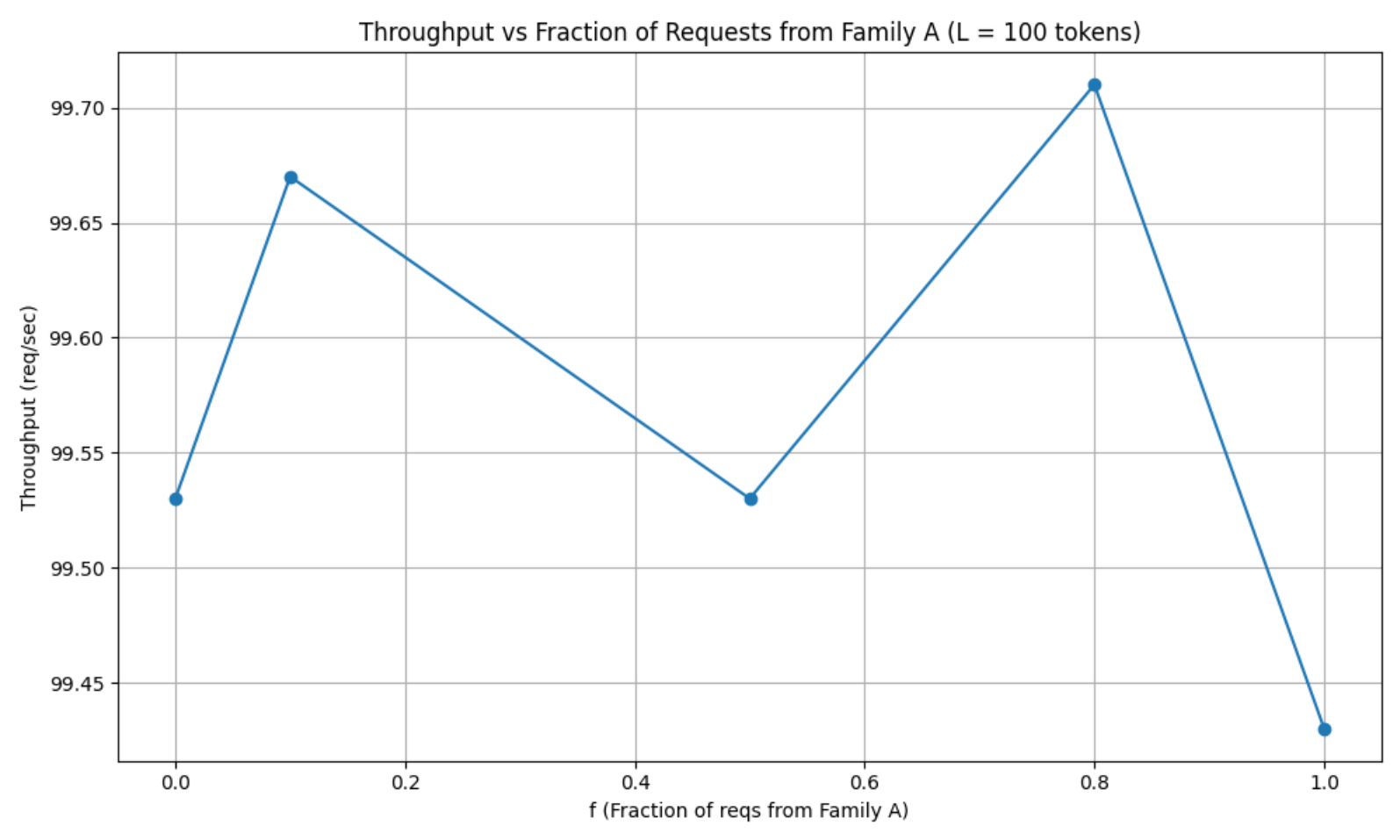


Figure: Results across fraction of requests from A

Two Family Batch Mixture (L = 500 tokens)

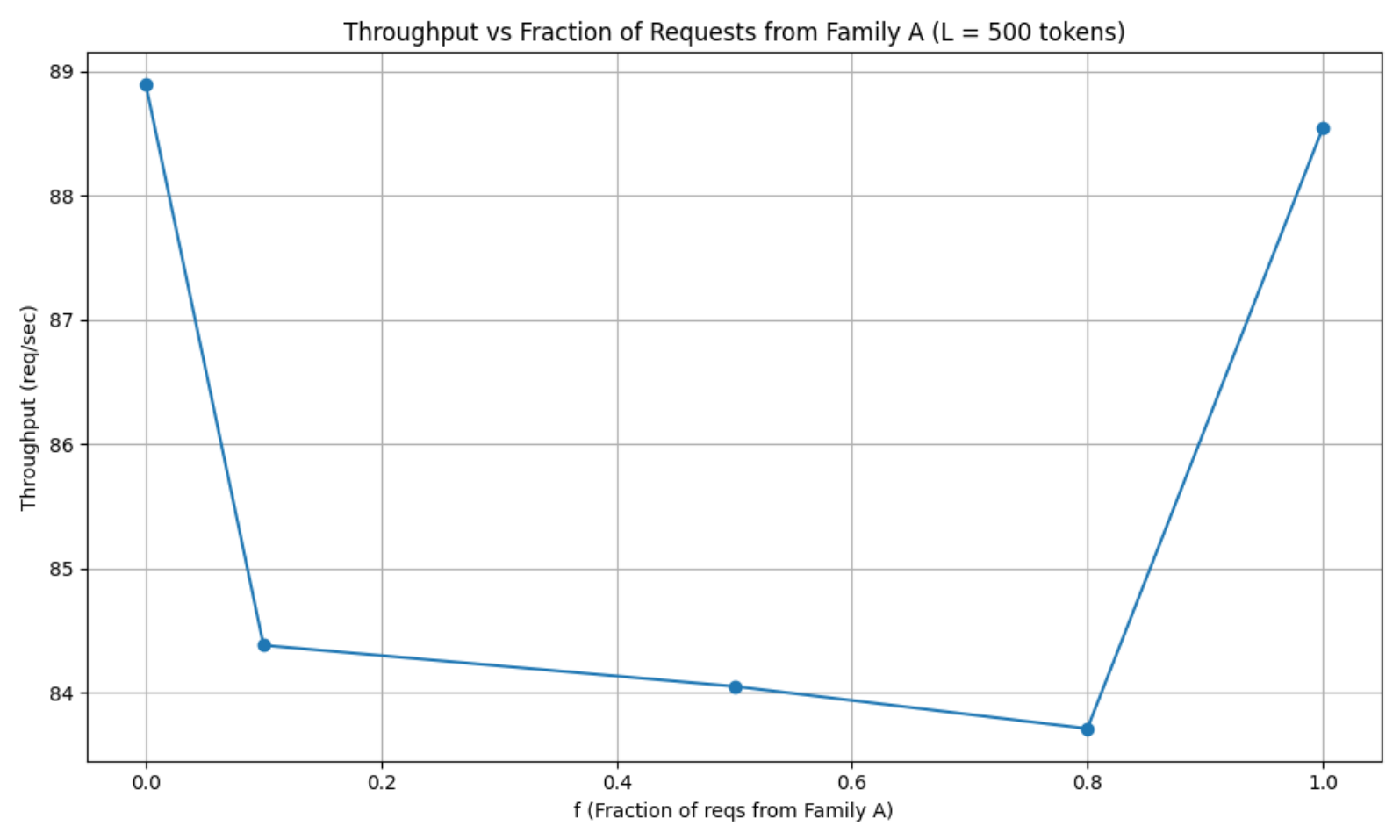


Figure: Results across fraction of requests from A

Two Family Batch Mixture (L = 1000 tokens)

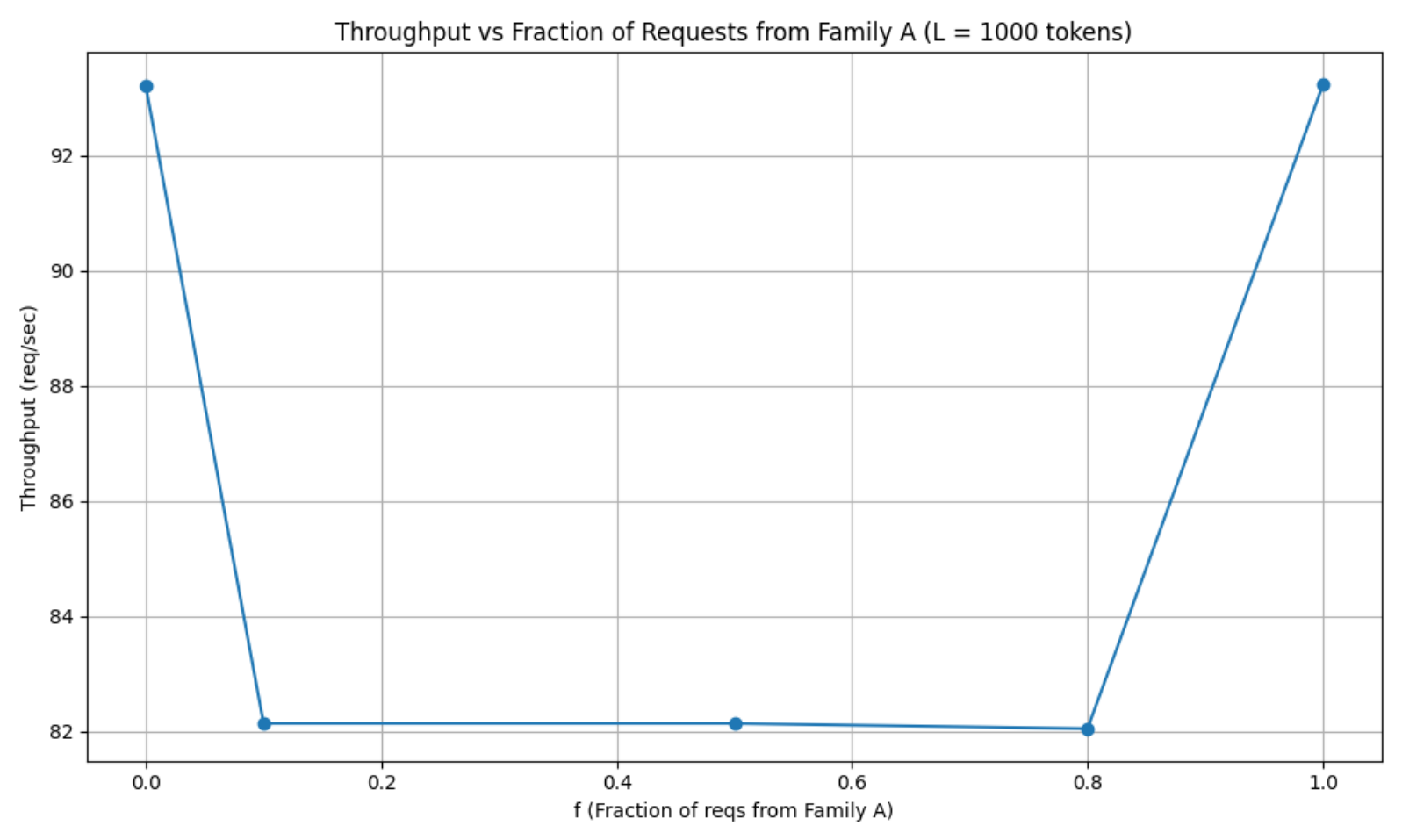


Figure: Results across fraction of requests from A

Two Family Batch Mixture (L = 2000 tokens)

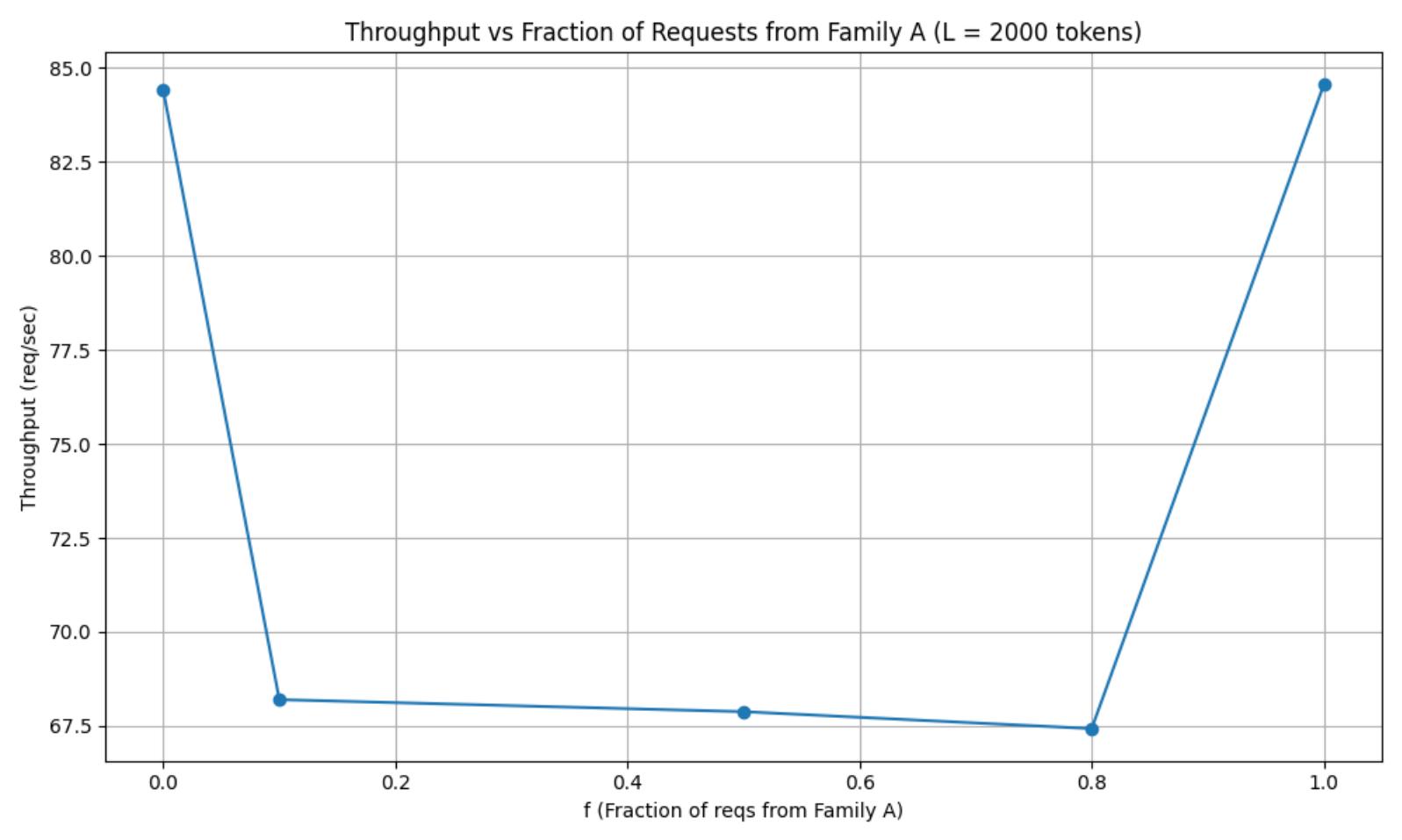
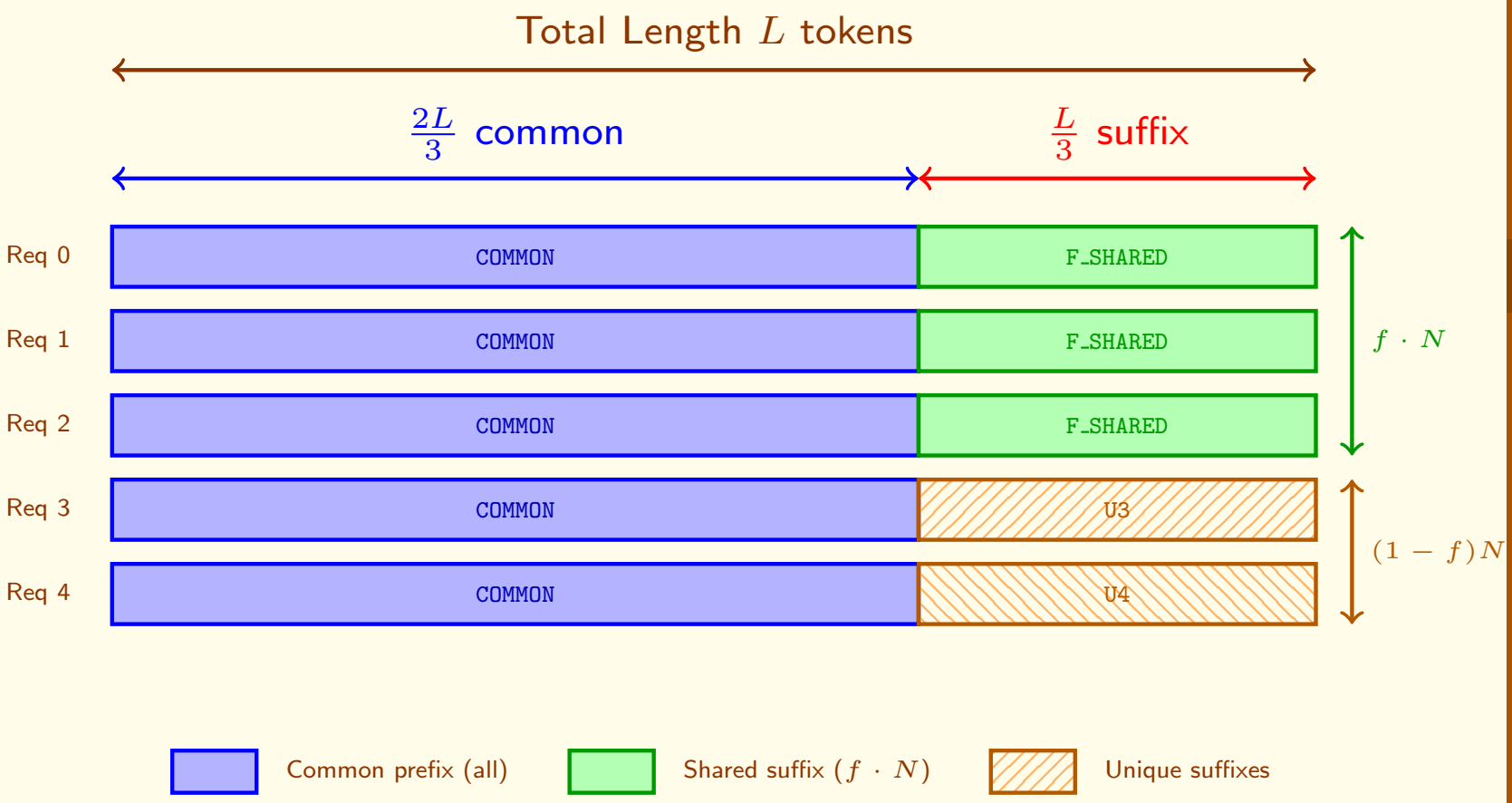


Figure: Results across fraction of requests from A

Tiered prefix sharing



Tiered Prefix Sharing (L = 500 tokens)

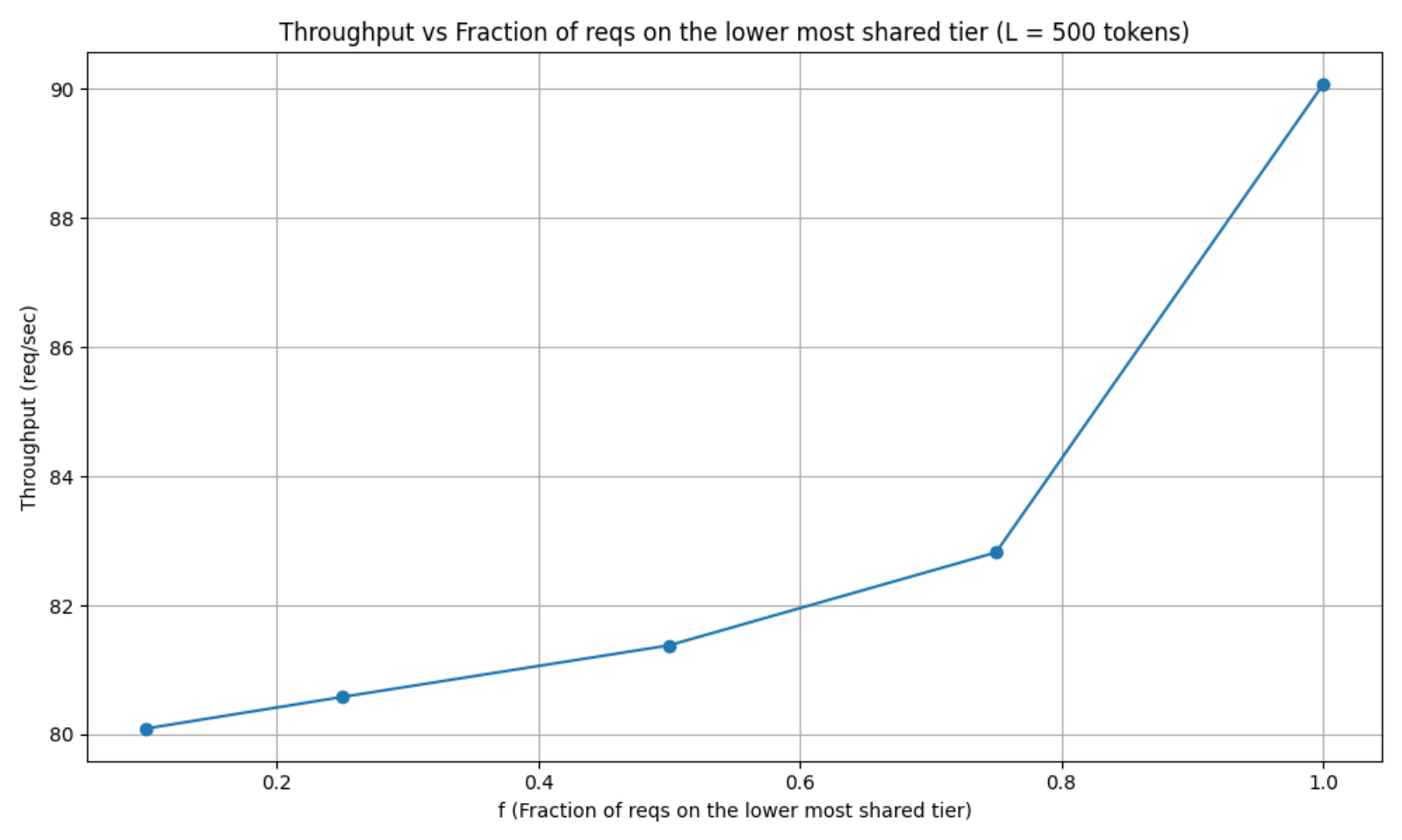


Figure: Varying fraction on lowermost tier

Tiered Prefix Sharing (L = 2000 tokens)

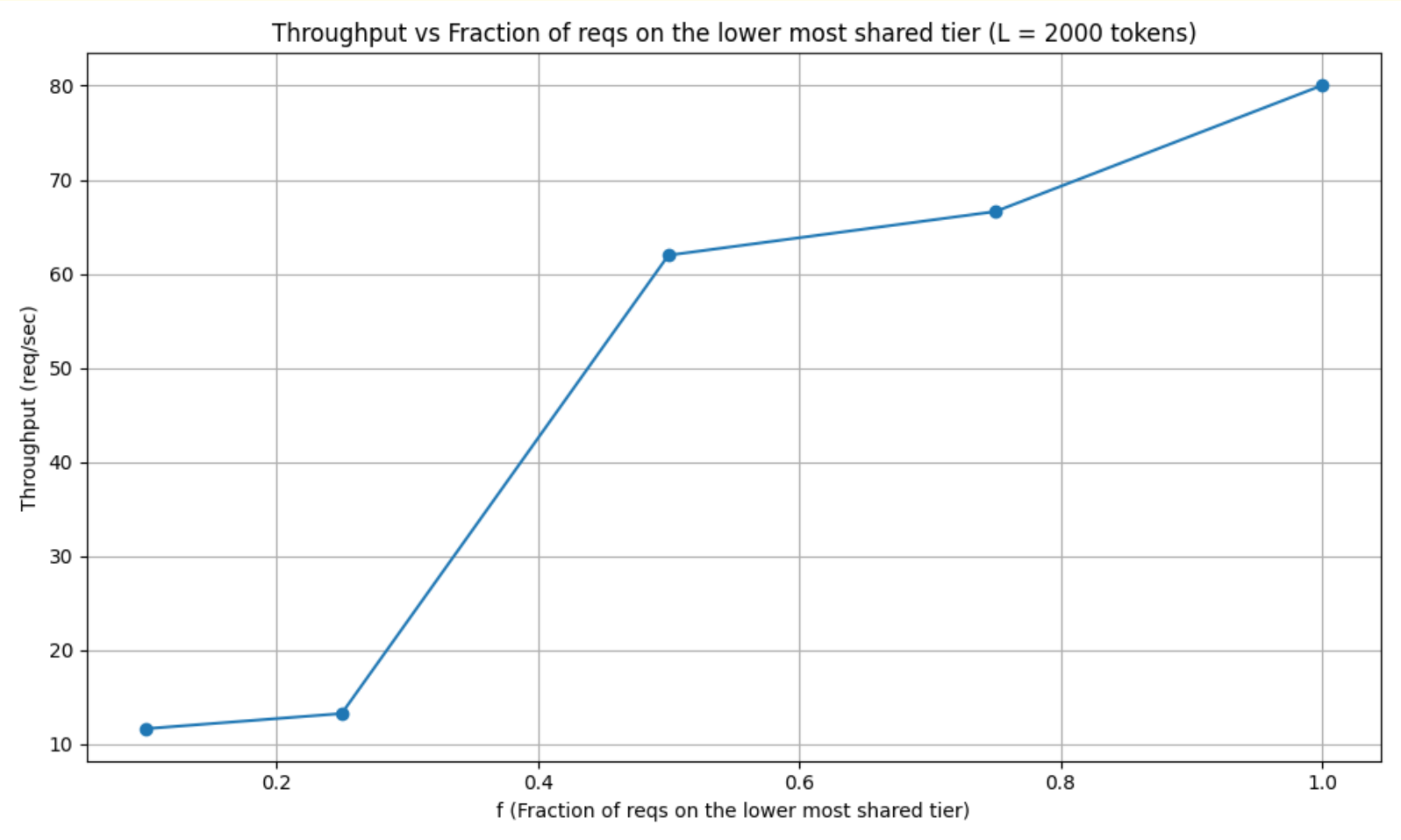


Figure: Varying fraction on lowermost tier

Throughput vs Max Batch Size

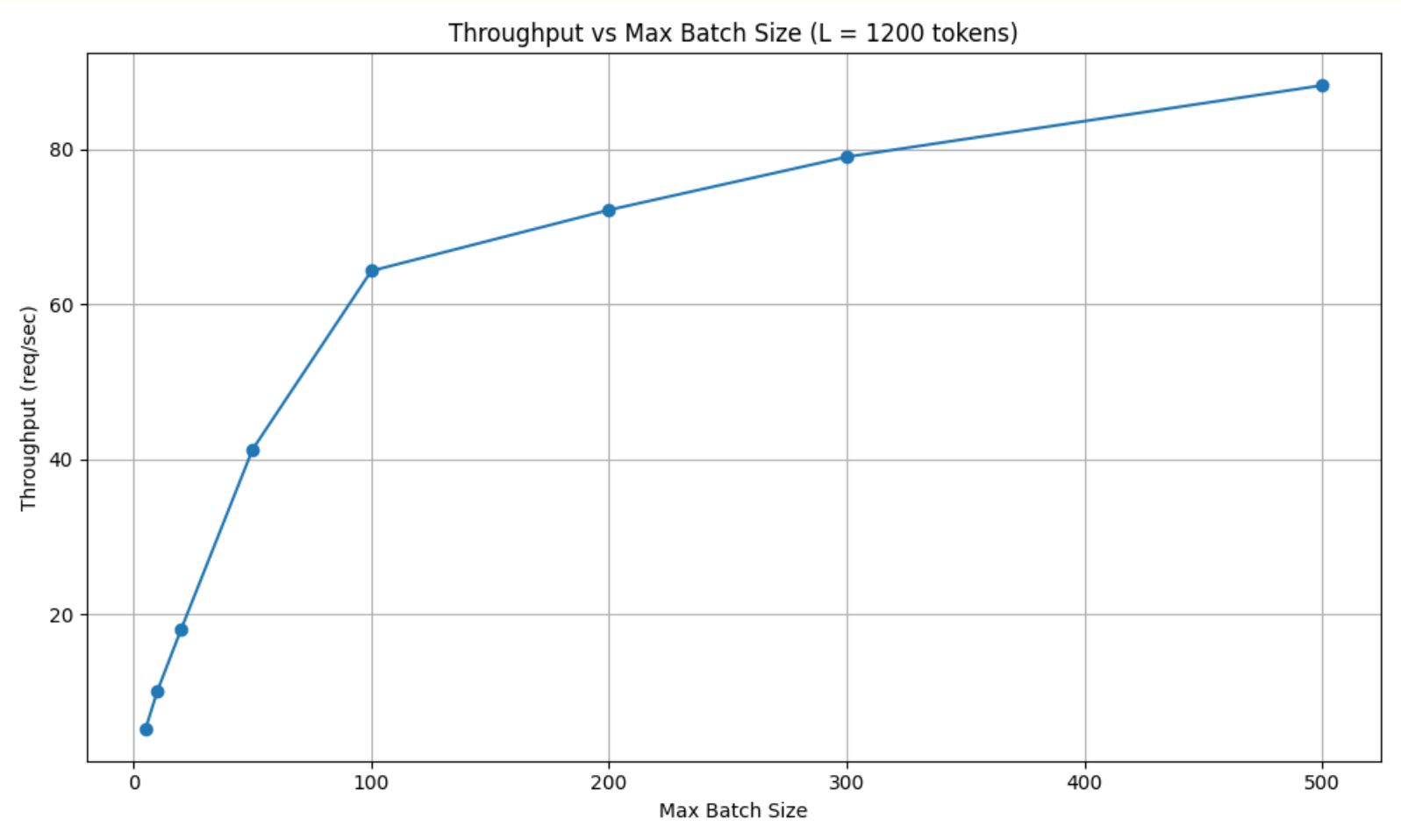


Figure: Throughput vs Max Batch Size (L = 1200 tokens)

Two Family Mixture - Mean DRAM Activity

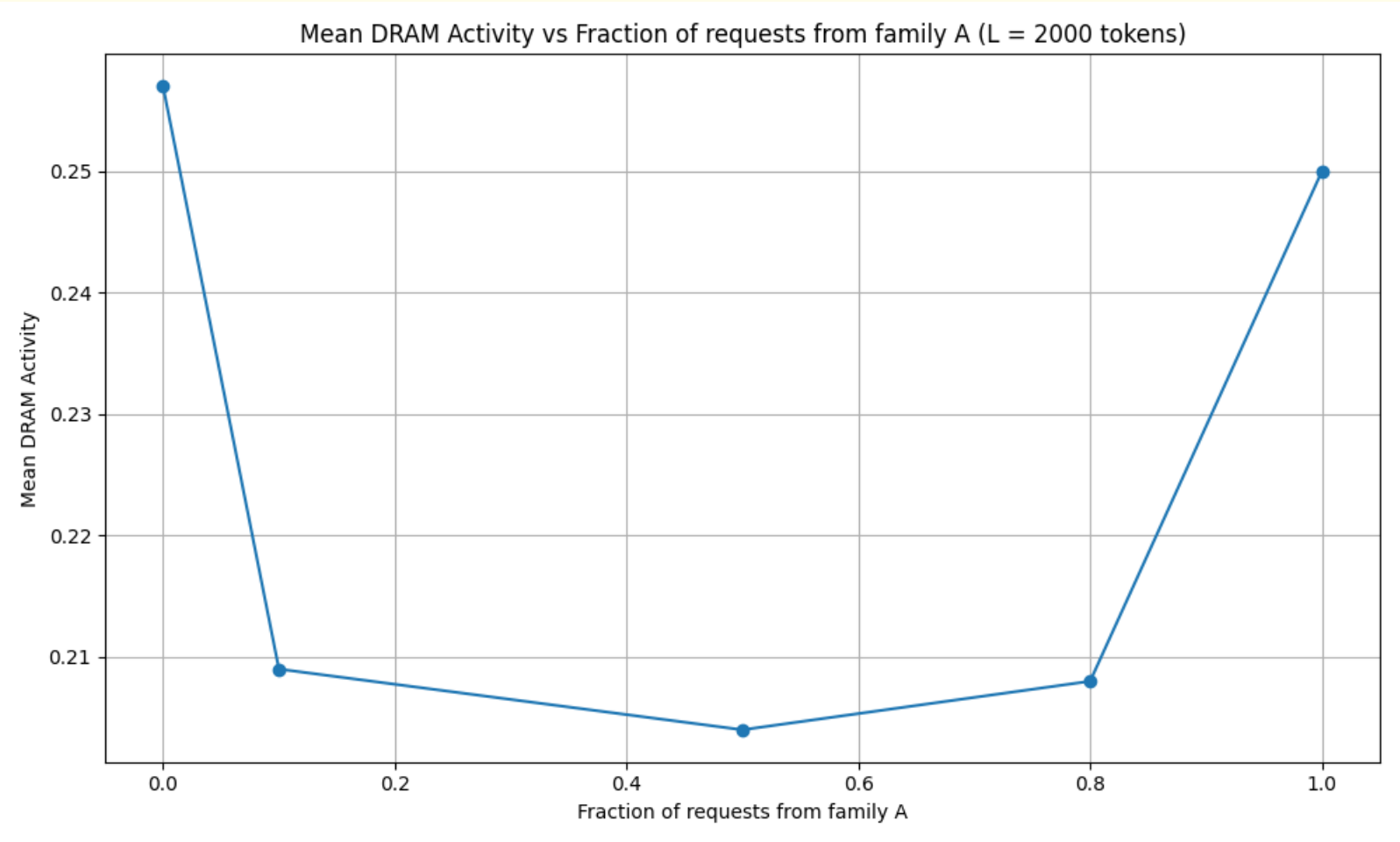


Figure: Mean DRAM Activity across fraction of requests from A (L = 2000 tokens)

Two Family Mixture - Total DRAM Accesses

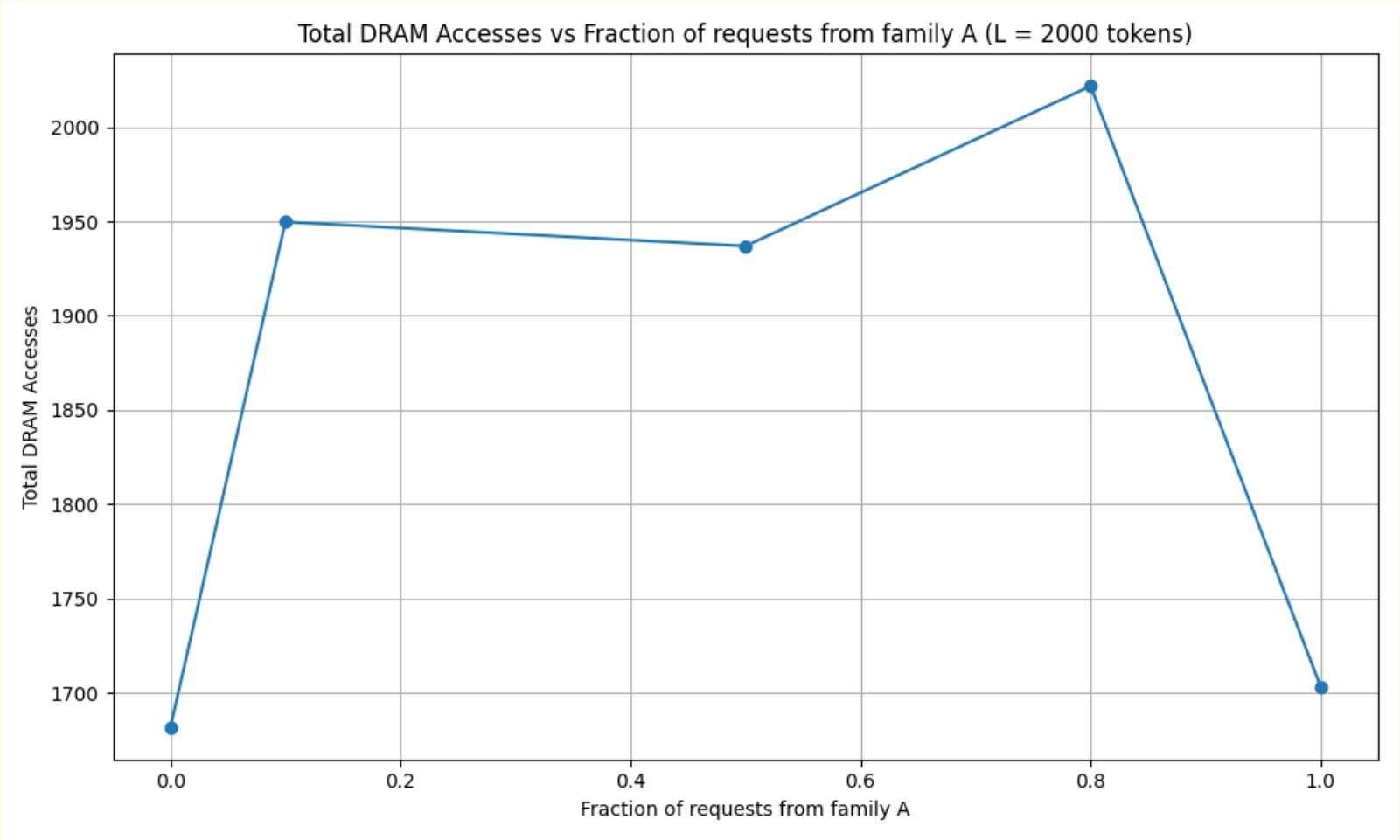


Figure: Total DRAM Accesses across fraction of requests from A (L = 2000 tokens)

Two Family Mixture - Mean SM Activity

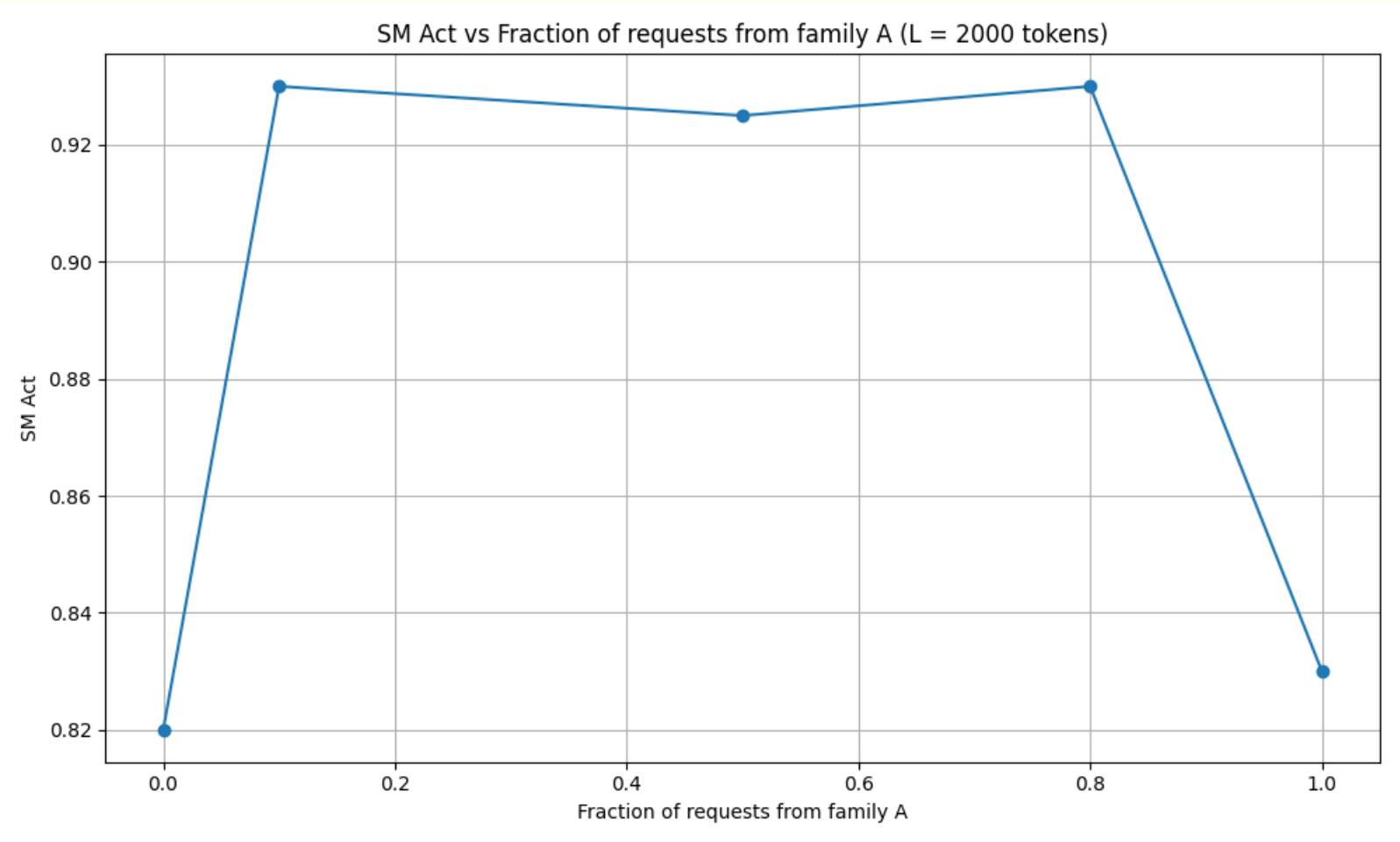


Figure: Mean SM Activity across fraction of requests from A (L = 2000 tokens)

Two Family Mixture - Mean SM Occupancy

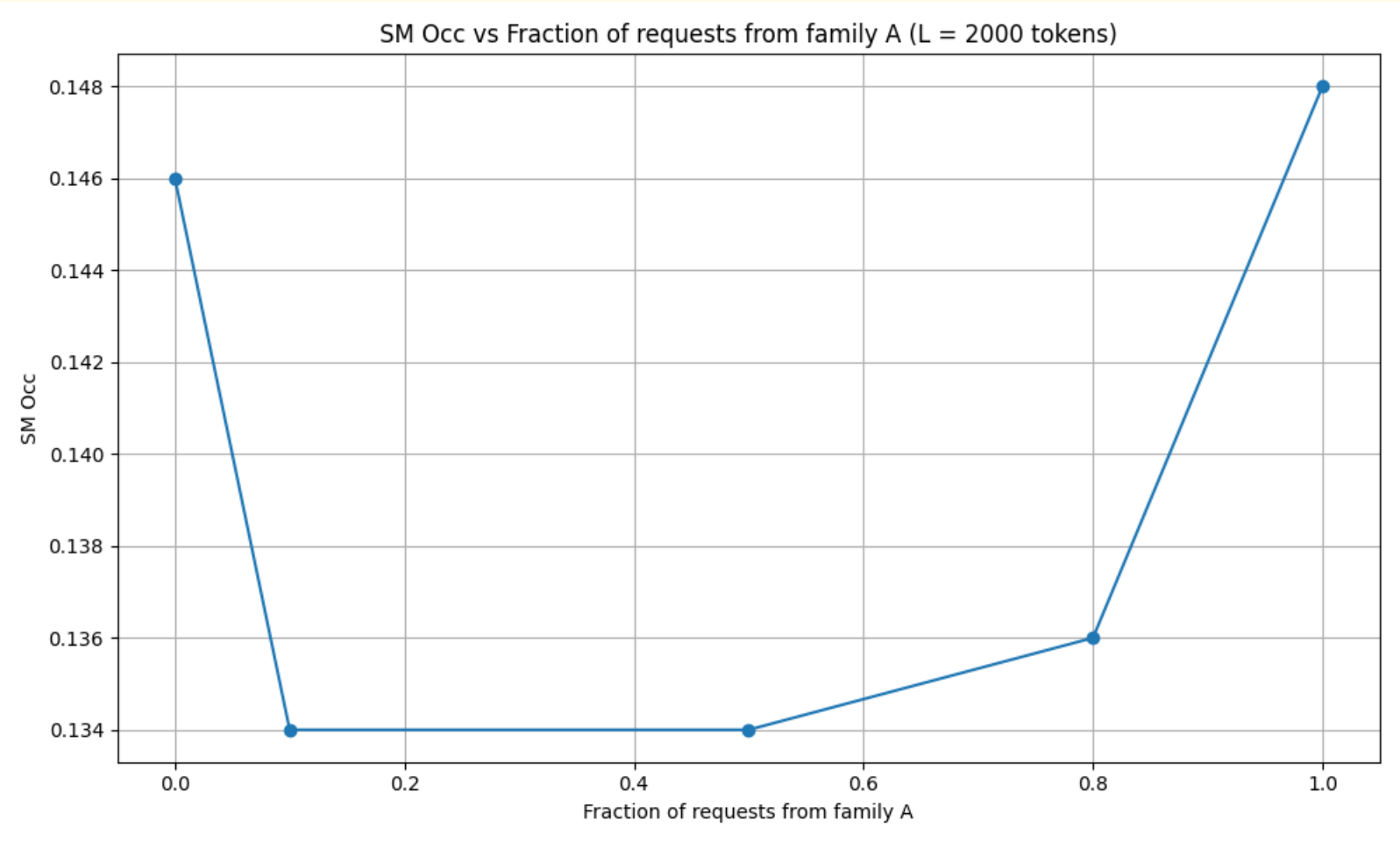


Figure: Mean SM Occupancy across fraction of requests from A (L = 2000 tokens)

Two Family Mixture - Mean Tensor Core Utilization

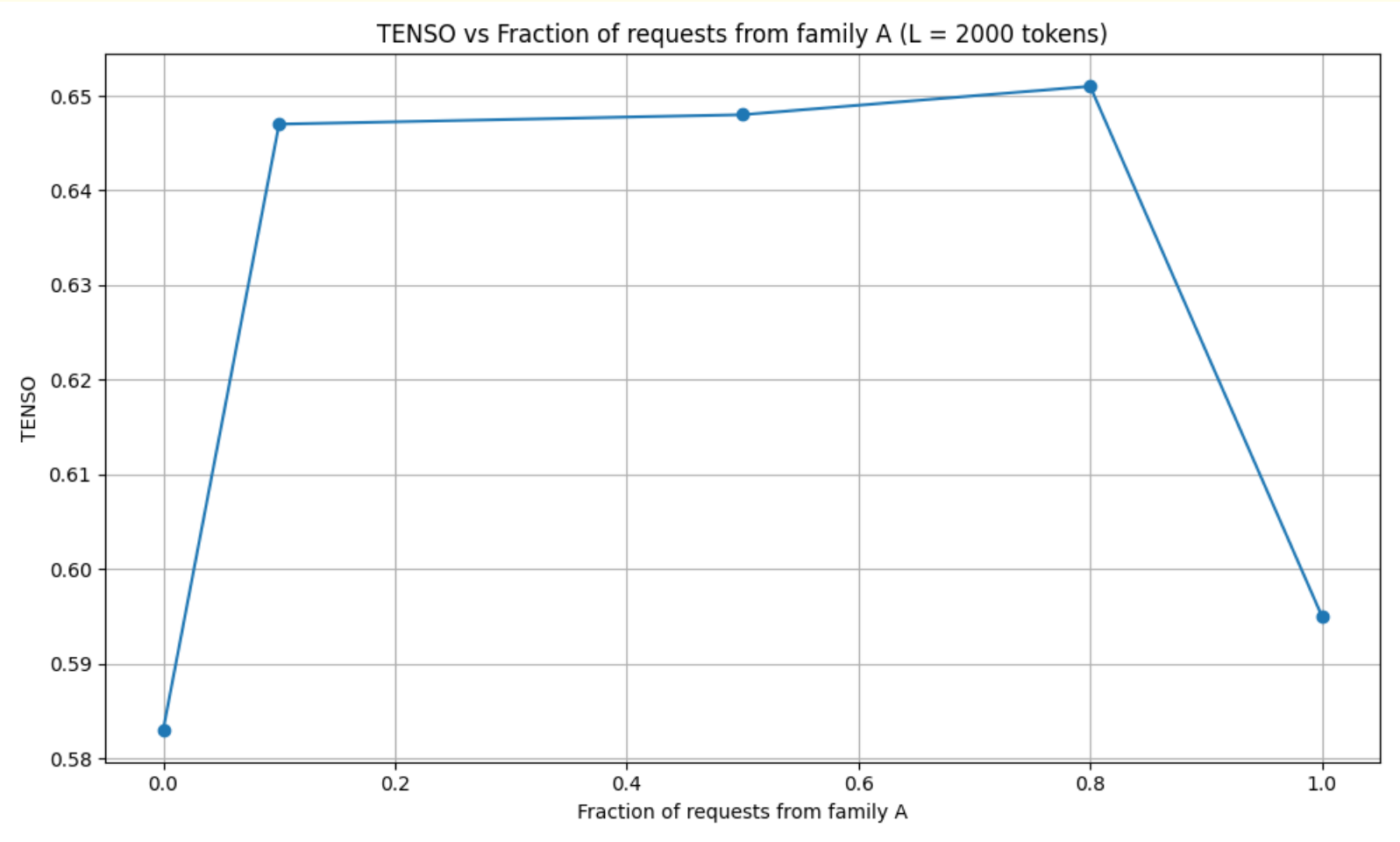
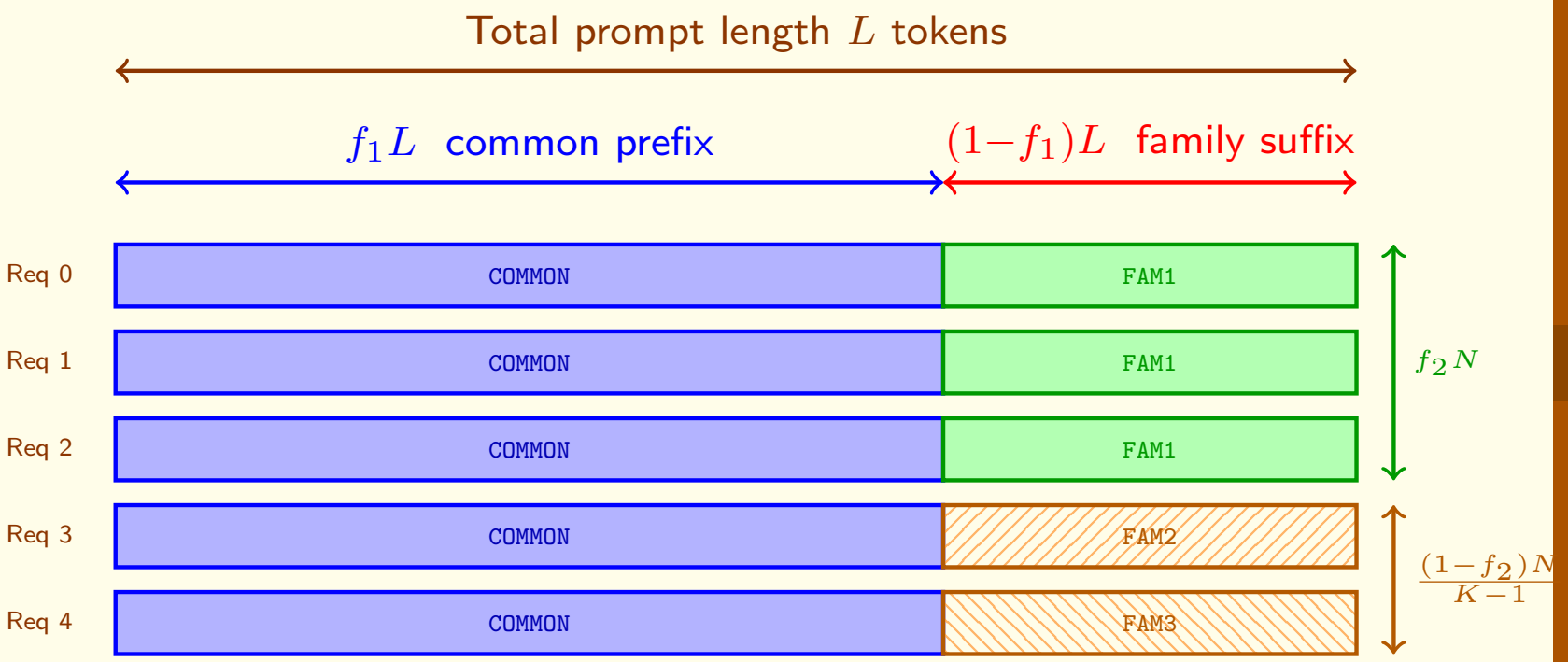


Figure: Mean Tensor Core Utilization across fraction of requests from A (L = 2000 tokens)

Tiered Prefix Sharing

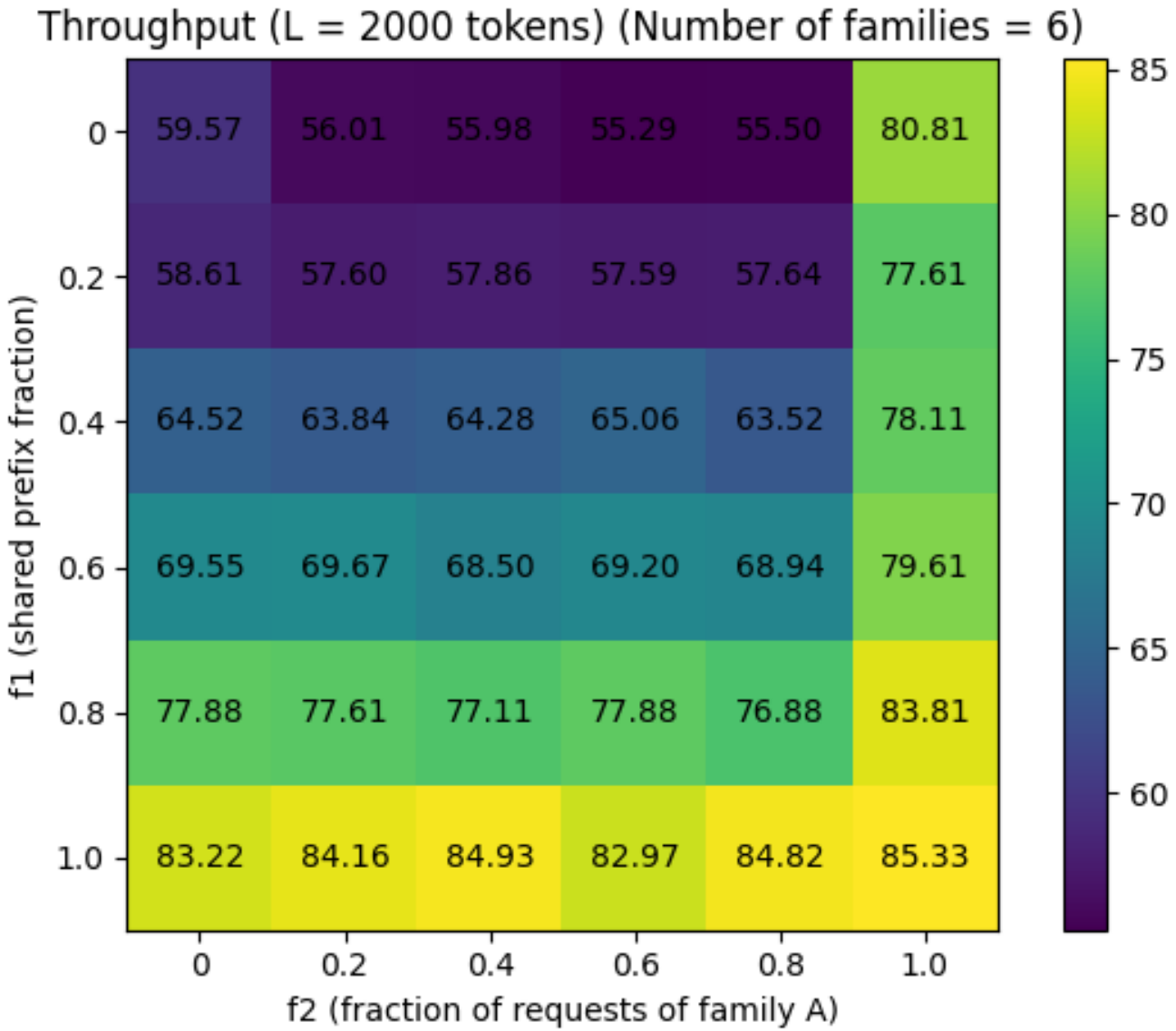


Common prefix — shared by *all* N requests ($f_1 L$ tokens)

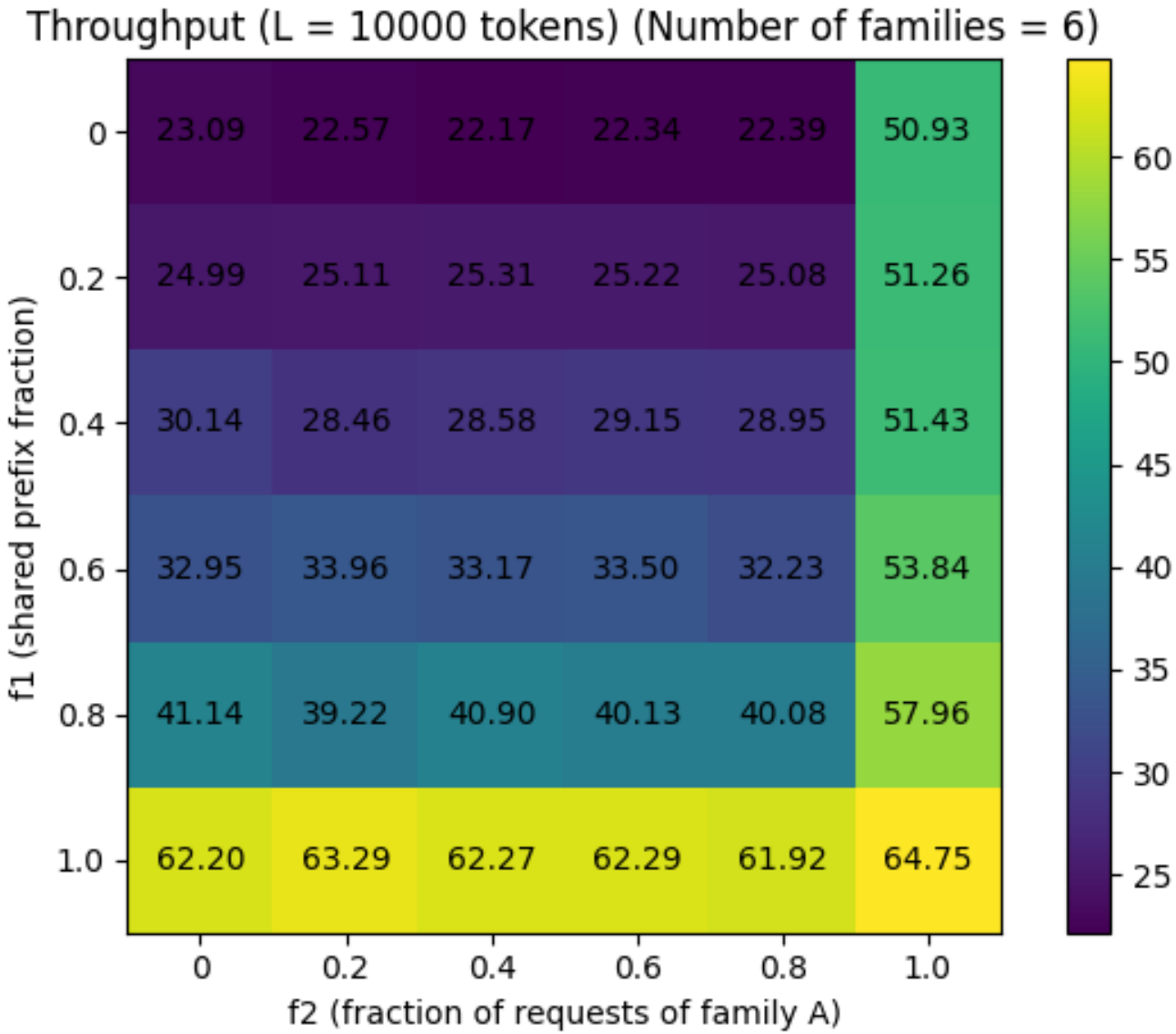
Family-1 suffix — shared by $f_2 N$ requests (dominant family)

Unique family suffix — $K-1$ minority families, $\frac{(1-f_2)N}{K-1}$ reqs each

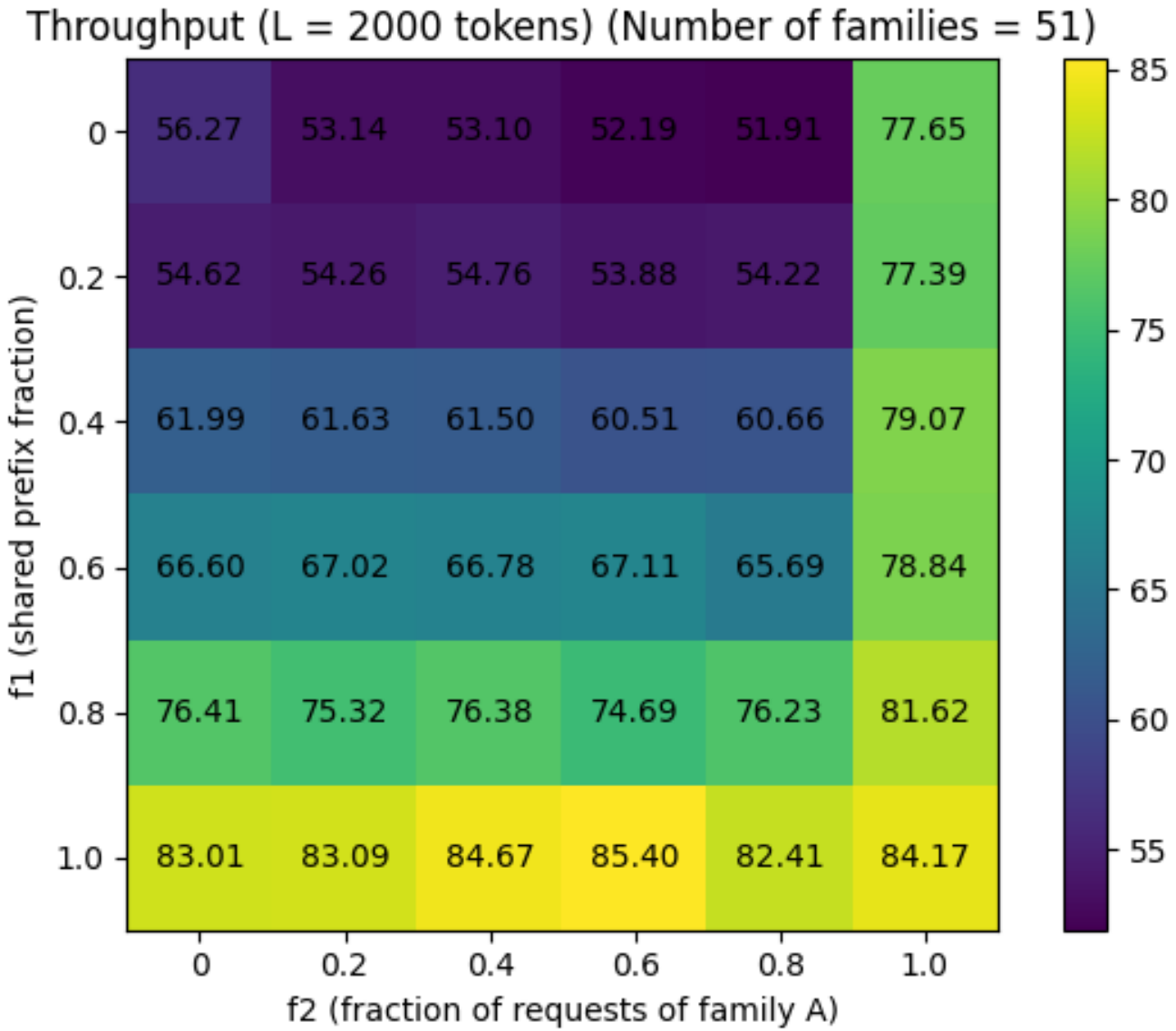
Tiered Prefix Sharing (L = 2000 tokens) (Number of families = 6)



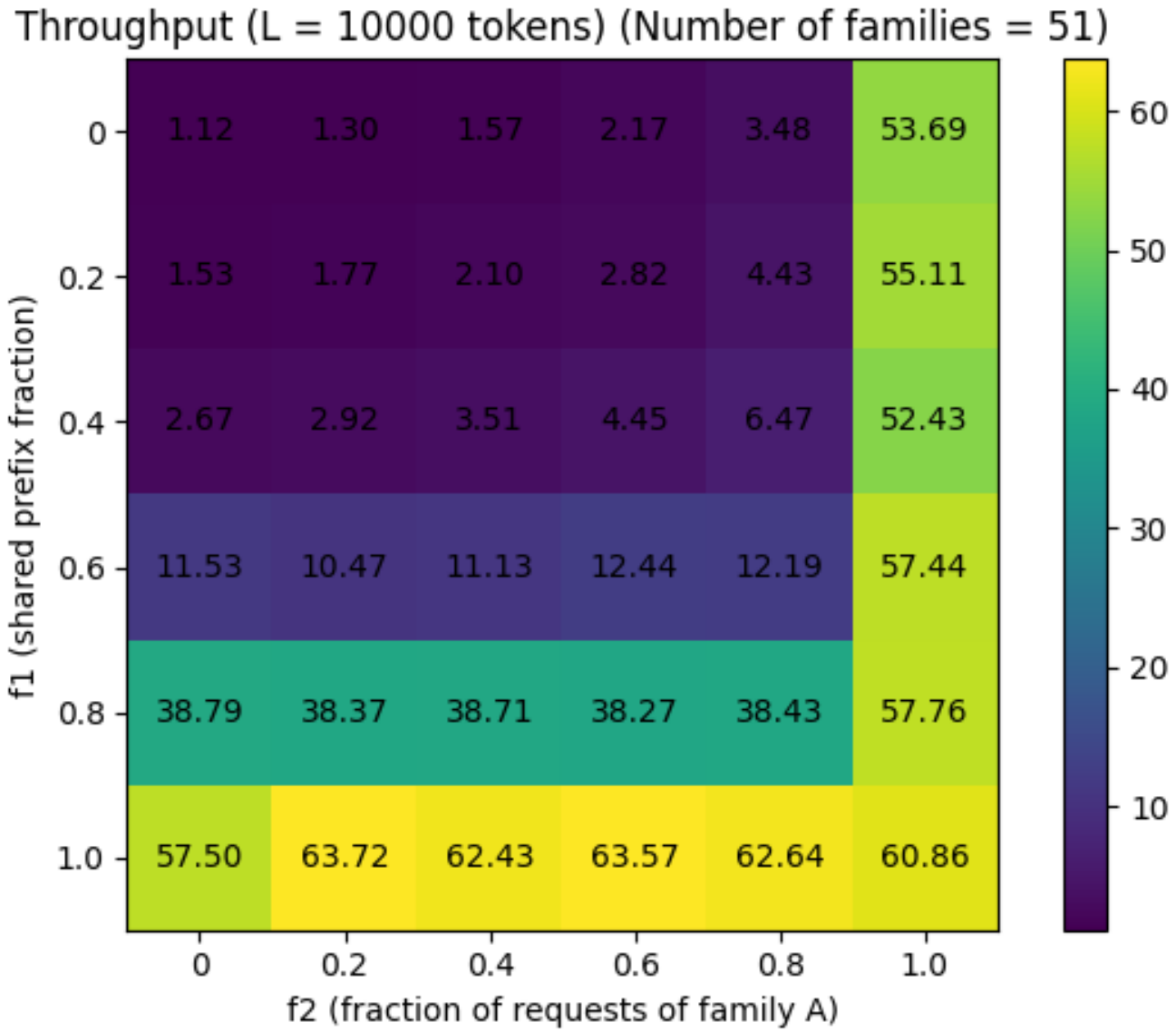
Tiered Prefix Sharing (L = 10000 tokens) (Number of families = 6)



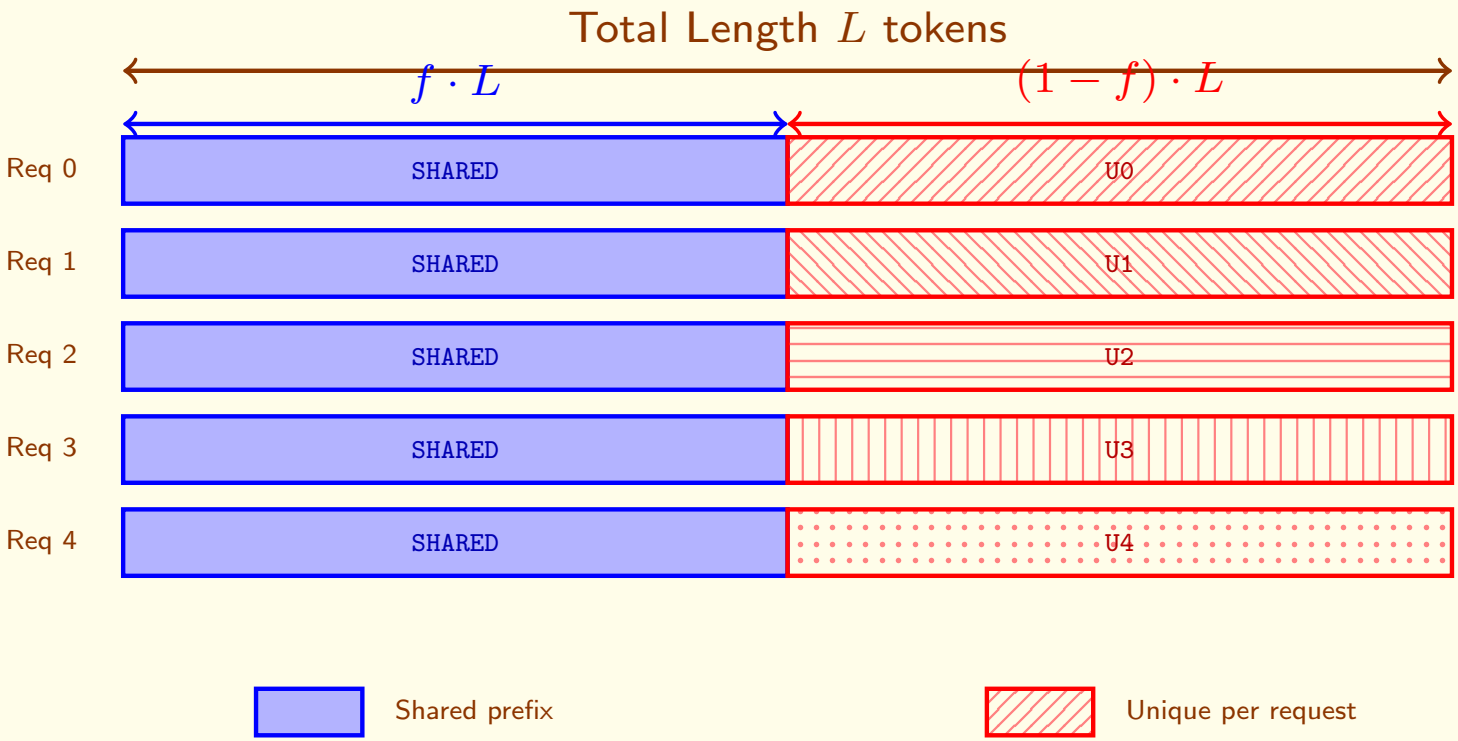
Tiered Prefix Sharing (L = 2000 tokens) (Number of families = 51)



Tiered Prefix Sharing (L = 10000 tokens) (Number of families = 51)



Varying the amount of sharing



Varying the amount of sharing - different RL Techniques

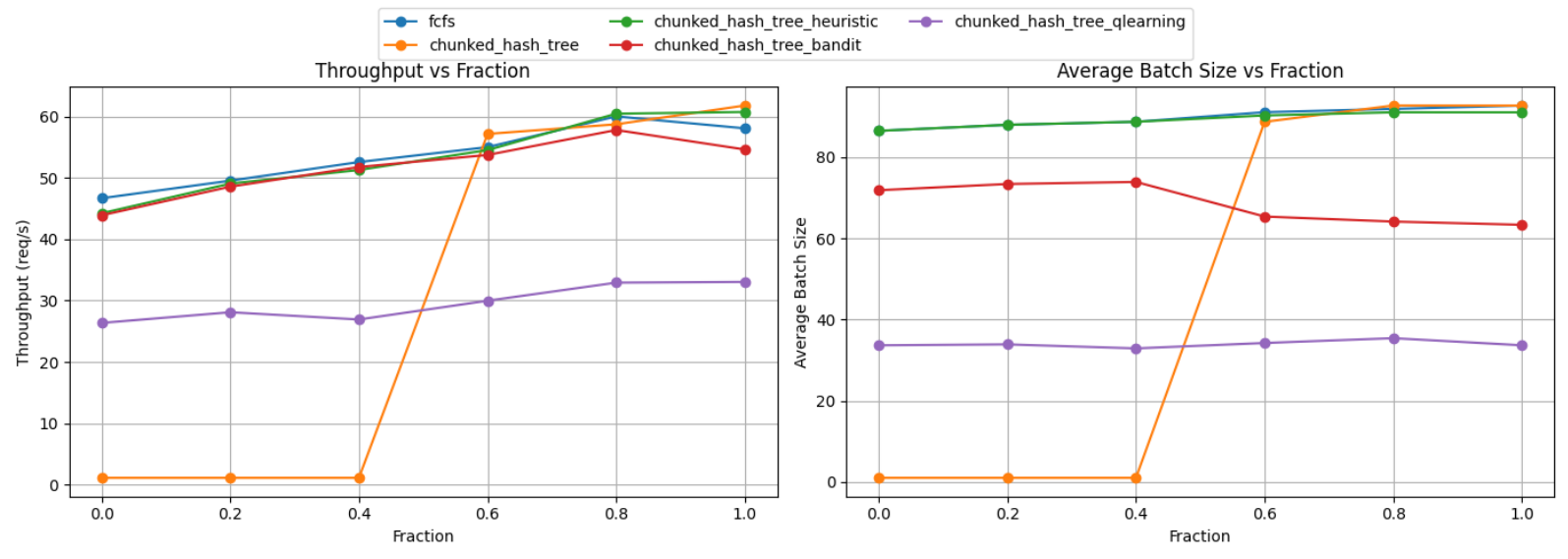


Figure: Throughput and Average Batch Sizes across Fractions ($L = 1200$ tokens, batch of 100 requests)



Thank You